

सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उदयम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

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Subject: 18th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

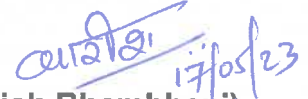
Dear Sir/Ma'am,

Please find enclosed the minutes of the 18th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 28th April, 2023 (Friday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,



(Kashish Bhambhani)
General Manager (CTU)

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Minutes of 18th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 28.04.2023

The 18th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 28/04/2023. List of participants is enclosed at **Annexure-X**.

A. Application related matters in Northern Region

1.0 Treatment of St-I, St-II, MTOA & LTA Applications received in Connectivity Regulations, 2009

- It was deliberated that Hon'ble CERC vide notification dated 01/04/2023 has issued First Amendment of GNA Regulations and notified that the effective date of Regulation 37.1 to 37.8 shall be 05/04/2023 in place of 15/10/2022.
- Further, under "Arrangement for Transition ", Reg. 37.1, it has been mentioned that "If an application for grant of Connectivity or grant of Long term Access or grant of Medium Term Open Access has been made in accordance with the Connectivity Regulations, 2009 and the same is yet to be granted as on the date of coming into effect of these Regulations, the applicant shall have the option of, either (a) to withdraw the application, in which case the application fee and bank guarantee, if any, shall be returned, or (b) to convert the application as an application made under these regulations by complying with the requirements under these regulations, which shall be processed in accordance with these regulations". Provided that such option shall be exercised by the applicant within one month of coming into effect of these Regulations, failing which the application shall be closed and the application fee and bank guarantee, if any, shall be returned.
- Only fresh requests received in this regard shall be considered for the purpose of transition under the GNA Regulations. In case no response is received by 04/05/2023, the option chosen shall be considered as surrendered.
- It was also deliberated that it has been notified on CTU website that Stage-I connectivity applicants/grantees, who have not applied for stage-II connectivity by 5th April 2023, may apply afresh as per GNA regulation along with applicable fees. Provided that original application fees submitted earlier by Stage-I connectivity applicants/grantees shall be refunded, if the fresh application for connectivity under GNA regulations is submitted by 4th May 2023. In addition to this, Inter-se priority of such applicants/grantees shall be determined with reference to their date of above fresh application under GNA Regulations.
- Further, it was noted that following St-I, St-II, LTA & MTOA applications at Table-1,2 & 3 respectively were received in the month of March'23 and April'23(till 04/04/23) in line with the Connectivity Regulations, 2009. In addition to this, applications received after 05/04/2023 under reverted category are also tabulated below:

TABLE 1

Sl. No.	Application ID	Application Category	Applicant	Project Location	Date of Application	Connectivity Sought Date	Quantum (MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity
1.	0112100062	St-I	Proteus Energy Private Limited	Bikaner, Rajasthan	10-03-2023	31/03/2025	300	Renewable Power Park Developer (Solar)	Bikaner-III PS
2.	0212100042	St-II	Proteus Energy Private Limited	Bikaner, Rajasthan	10-03-2023	31/03/2025	217 (L&FC)	Renewable Power Park Developer (Solar)	Bikaner-III PS
3.	0212100043	St-II	Tepsol Saur Urja Private Limited	Barmer, Rajasthan	31-03-2023	31-12-2025	300 (L&FC)	Renewable Power Park Developer (Solar)	Fatehgarh-IV PS
4.	0112100068	St-I	Rajasthan Solarpark Development Company Limited	Bikaner, Rajasthan	07-04-2023	30/03/2024	450	Renewable Power Park Developer (Solar)	Bikaner-III PS
5.	0112100070	St-I	Rajasthan Solarpark Development Company Limited	Bikaner, Rajasthan	12-04-2023	23/02/2024	1000	Renewable Power Park Developer (Solar)	Bikaner-III PS
6.	0312100021	St-II (Enhancement)	Juniper Green Beta Private Limited	Jodhpur, Rajasthan	12-04-2023	30/06/2025	40	Generator (Solar)	Bhadla-III PS
7.	0312100019	St-II	Cannice Renewables Energy Private Limited	Barmer, Rajasthan	12-04-2023	31-01-2025	90	Generator (Hybrid) [Solar: 70MW, Wind: 20MW]	Fatehgarh-IV PS

TABLE 2

Sl. No.	Application No.	Applicant	LTA Application date (online)	Connectivity Injection Point	Drawl Point	Quantum of LTA (MW)	Start Date of LTA	End date of LTA
1	0412100014	Serentica Renewables India Private Limited	17-03-2023	Bikaner-II PS (NR)	Kankroli (PG) S/s	180	31-07-2024	30-07-2049
2	0413600001	THDC India Limited (Khurja STPP)	24-03-2023	Aligarh S/s, UP (NR)	JVVNL, AVVNL, JdVVNL: 281.1MW UPCL: 51.8MW Unallocated: 132.7MW	465.6	30-04-2024	29-04-2049
3	0412100027	Juniper Green Beta Private Limited	12-04-2023	Bhadla-III PS (NR)	Target (NR)	50	31-03-2026	31-03-2051

TABLE 3

Sl. No.	Application ID	Name of the Applicant	MTOA Application date (online)	Drawl Region	Quantum of MTOA	Start Date of MTOA	End Date of MTOA	Injection Point	Drawl Point
1.	0513600001	Tata Power Trading Company Limited	17-03-2023	NR	100.0	01-09-2023	31-03-2026	JITPL, Odisha, ER	Noida Power Company Limited, Uttar Pradesh (NR)
2.	0512100003	JSW Neo Energy Limited	30-03-2023	NR	100.0	01-09-2023	31-03-2024	Tuticorin-II PS, Tamil Nadu (SR)	RRVPL, Rajasthan (NR)
3.	0512100004	JSW Neo Energy Limited	04-04-2023	NR	100.0	01-10-2023	31-03-2024	Tuticorin-II PS, Tamil Nadu (SR)	RRVPL, Rajasthan (NR)
4.	0512100005	JSW Neo Energy Limited	04-04-2023	NR	100.0	01-10-2023	31-03-2024	Tuticorin-II PS, Tamil Nadu (SR)	RRVPL, Rajasthan (NR)

- Further, for St-I, St-II & LTA applications received under reverted category in the month of April'23, in line with the Connectivity Regulations, 2009, it was informed that the priority for such applications shall be maintained with reference to the other applications received under GNA Regulations also as per respective application date. However, GNA transition offer has already been intimated to these applicants (St-II & LTA) as well. Applicants noted the same.
- Accordingly, it was deliberated that all concerned have to take expeditious action towards exercising option to convert/surrender their respective quantum(s), as applicable, for transition under the GNA Regulations through letter/applicable application FORMAT available on CTU website in offline mode on or before 04/05/2023.
- For MTOA application (Sl. No.1: Table 3) from M/s Tata Power Trading Company Limited for 100 MW, it was informed that as per Reg 37.5 under GNA Regulations'23(first amendment), the MTOA/LTA granted under connectivity regulation 2009 which has not become effective, to a trading licensee, other than that covered under Regulation 17.1 of these regulations shall be treated as withdrawn and BG furnished, if any shall be returned. In view of this, the transition offer is not been sent to M/s Tata Power. Applicant noted the same and informed that they will withdraw their MTOA application.
- Further, various queries from generation developers were taken up by CTUIL and it was concluded that applications which are under processing may opt/surrender on or before 04/05/2023 and in case no response is received by 04/05/2023, the option chosen shall be considered as surrendered. All applicants noted the same.

B. ISTS Expansion

1.0 Transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) & Tidong HEP in Himachal Pradesh

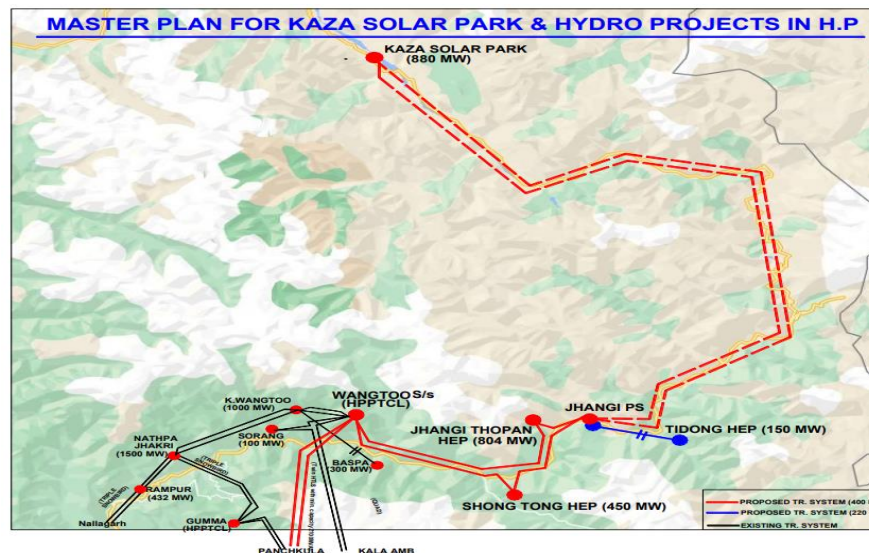
It was informed that a comprehensive transmission system (400 kV Jhangi-Wangtoo-Panchkula D/c Corridor) for evacuation of power from two Hydro Electric Projects (HEPs) viz Tidong (150 MW) and Shongtong Karcham HEP (450 MW) of HPPCL & Tidong Power Generation Private Limited respectively in Himachal Pradesh was agreed in the 10th Consultation Meeting for Evolving Transmission Schemes in Northern Region meeting held on 30/08/2022. Above transmission system was also agreed in 57th Northern Region Power Committee (NRPC) meeting held on 31/08/2022 & 11th National Committee on Transmission (NCT) held on 28/12/2022 & 17/01/2023 with following broad scope of works to be implemented in 2 phases with Tidong HEP system in Ph-1 & Shongtong HEP system in Ph-2:

Phase-I (with Tidong HEP and with schedule: 01st July 2026)

- Establishment of 2x315 MVA (7x105 MVA), 400/220 kV GIS Pooling Station at Jhangi
- Jhangi PS – Wangtoo (Quad) 400 kV D/c line
- 2 nos. of 400 kV bays at Wangtoo (HPPTCL) for termination of 400 kV Jhangi PS – Wangtoo D/c line

Phase-II (with Shongtong HEP and with schedule: 31st July 2026]

- LILO of one circuit of Jhangi PS – Wangtoo (HPPTCL) 400 kV D/c (Quad) line at generation switchyard of Shongtong HEP
- Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line (Twin HTLS)
- 400 kV bays at Wangtoo S/s (2 nos.) and Panchkula S/s (2 nos.) for termination of Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line



The transmission scheme was also approved by MOP based on recommendation by the NCT for implementation through TBCB route. Recently, HPPCL vide their letter dated 05.04.2023 intimated that the commissioning date of Shongtong Karcham HEP (STKHEP) had been revised (preponed) from July'26 to July'25 and requested to review the transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) in Himachal Pradesh due to the preponement in the commissioning timeline of STKHEP.

Accordingly, a meeting was convened by CEA on 12.04.2023 regarding the above development in the generation schedule of Shongtong HEP(**copy of minutes enclosed in Annexure-I**). During the meeting, HPPCL informed that the progress of the STKHEP was reviewed and the various issues/disputes that were delaying the project were resolved and therefore their revised commissioning schedule shall be now July'25. M/s Statkraft (for Tidong HEP) stated that for Tidong HEP, the requirement of Jhangi PS would be from July'26, which was mentioned in their Connectivity application. M/s Statkraft added that since the location of Jhangi Pooling station has not yet been finalized and considering the issue of tough terrain and local RoW issues, constructing dedicated line from Tidong HEP to Jhangi Pooling station would take considerable time and hence, the requirement of transmission system for Tidong HEP would be from July'26.

CTUIL during the meeting informed that these are very high altitude areas with long periods of no activity in winter and monsoon season and considering the various difficulties in construction of transmission line in that area, the completion of the transmission system in July 2025 timeframe does not seem to be practical. Further, the implementation schedule of the transmission system to be implemented in challenging areas like hilly terrain may be taken as 30/36 months. CEA agreed and stated that the minimum timeframe for hilly terrain is to be considered as 30-36 months. HPPCL stated that from the total length of 54 kms of the 400 kV transmission line from Jhangi PS to Wangtoo (HPPTCL), the 18 km portion from Wangtoo (HPPTCL) to Shongtong HEP could be implemented by July 2025, which would be sufficient for evacuation of power from STKHEP till the completion of complete final transmission system.

On query from CEA about the bifurcation of the transmission scheme in two phases, CTUIL stated that the 18 km portion of the line could be considered as an interim arrangement; however, the power transfer from STKHEP could not be committed and would be on real time basis depending on the available margins in the system. STKHEP may have to apply for T-GNA for evacuation of power, till the Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line is implemented.

In view of above, it was decided that in the case of different timelines for the requirement of transmission system for Tidong HEP and STKHEP, the transmission system could be phased accordingly. Further schedule for interim part was decided to be 24 months.

After further deliberation, in above meeting, it was agreed that transmission system to be bifurcated in two parts as mentioned below:

(i) Interim Part (with time frame of 24 months)

- Generation switchyard of Shongtong HEP to Wangtoo (HPPTCL) 400 kV D/c line (about 18 kms)
- 2 nos. of 400 kV bays at Wangtoo (HPPTCL)

(ii) Final System (with time frame of 1st July 2026)

- Establishment of 2x315 MVA (7x105 MVA 1-ph units including one spare unit), 400/220 kV GIS Pooling Station at Jhangi
- Completion of remaining portion of 400 kV line i.e. from Shongtong HEP to Jhangi PS so as to form Jhangi PS – Wangtoo (Quad) D/c line with LILO of one circuit of Jhangi PS – Wangtoo (HPPTCL) 400 kV D/c (Quad) line at generation switchyard of Shongtong HEP
- Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line
- 400 kV bays at Wangtoo S/s (2 nos.) and Panchkula S/s (2 nos.) for termination of 400 kV Wangtoo (HPPTCL) - Panchkula (PG) D/c line

The above scheme was also discussed in the recent 65th NRPC meeting held on 21/04/2023. During the NRPC meeting, HPPCL reiterated that their generation project shall be commissioned by July'25 and the implementation of 18 km portion of the ISTS line could be considered on urgent basis for evacuation of their generation in given timeline.

CTU in the meeting opined that a meeting was held on 25/11/22 wherein it was agreed that schedule of transmission system to be implemented in challenging areas like hilly terrain may take 30/36 months. MS (NRPC) stated that implementation of the interim arrangement with the time frame of July'25 matching with the generation project in such hilly terrain though difficult but not impossible. Therefore, they shall be recommending the NCT to give consideration to generation project schedule and accordingly line may be developed suitably.

NRPC suggested that scheme should also be posed to NR constituents in CMETS-NR meeting for their views. In NRPC meeting, CTU stated that conductor of 400kV Jhangi – Wangtoo D/c (one ckt via Shongtong HEP) may be noted as Quad (2500 MVA) and 400kV Wangtoo (HPPTCL) - Panchkula (PG) may be noted as Twin HTLS (2100 MVA), as per earlier approved scheme. Based on above, modified Transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) & Tidong HEP^{##} in Himachal Pradesh was proposed as under:

(i) Interim System [For Shongtong HEP: 24 months]

- Generation switchyard of Shongtong HEP^{**} to Wangtoo (HPPTCL) 400 kV D/c (Quad^{\$}) line (about 18 kms)
- 2 nos. of 400 kV bays (GIS) at Wangtoo (HPPTCL)

(ii) Final System (To be matching with generation schedule ie.1st July 2026)

- Establishment of 2x315 MVA (7x105 MVA 1-ph units including one spare unit), 400/220 kV GIS Pooling Station at Jhanggi [along with future provision]
- Extension of Wangtoo (HPPTCL) - Shongtong HEP 400 kV (Quad)^{\$} D/c line upto Jhanggi PS with one circuit through Shongtong HEP generation switchyard
- Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line (Twin HTLS)*
- 80 MVAR switchable line reactor at Panchkula end on each circuit of 400 kV Wangtoo (HPPTCL) - Panchkula (PG) D/c line
- 125 MVAR, 420kV Bus reactor at Jhanggi PS (1-ph units along with one spare unit)

\$ Line capacity shall be 2500 MVA per circuit at Nominal voltage

** with minimum capacity of 2100 MVA on each circuit at Nominal voltage*

*** Associated 400 kV bays at Shongtong Generation switchyard shall be under the scope of applicant/generation developer*

##Tidong HEP- Jhanggi PS 220 kV D/C line (along with associated bays at both ends) - under the scope of applicant/generation developer.

During, the present meeting, HPPCL reiterated that they shall commission their Shongtong generation by July'25 and the decision should be taken in line with the deliberations held in the NRPC meeting held on 21/04/2023.

Further, CTUIL highlighted that power from Shongtong HEP & Tidong HEP, who sought Connectivity & LTA in ISTS, was to be pooled at Wangtoo (HPPTCL) S/s. However, Wangtoo (GIS) S/s pertains to STU i.e. HPPTCL. Regarding this, in the 50th meeting of NR constituents regarding Connectivity & LTA Applications in NR` earlier, HPPTCL informed that considering issues in development of new substations in the terrain and availability of 4 nos. bays at existing Wangtoo S/s, requested to utilize the existing station and defer plan of new Wangtoo ISTS substation.

Further HPPTCL informed that they have already filed a petition in Hon'ble CERC for declaring Wangtoo 400/220 kV S/s as an ISTS asset, as it is envisaging the ISTS power of regional entities. Further, many of the hydro projects in Satluj basin area may seek connectivity through ISTS for which above proposed system incl. Wangtoo S/s shall be utilized. In view of this, it was agreed that as the 400 kV side of Wangtoo S/s shall be used for evacuation of power under ISTS, accordingly, HPPTCL may take up the matter with CERC for urgent hearing.

Further, M/s Statkraft during the meeting requested that implementation of 220 kV bays at Jhanggi PS end for Tidong HEP DTL may be taken up under ISTS under GNA regime. CTU informed that they will look into the requirement in light of the effectiveness of GNA regulation.

Accordingly, following revised scheme was deliberated and agreed in the CMETS-NR meeting:

(i) Interim System [For Shongtong HEP: with time frame of 24 months]

- Generation switchyard of Shongtong HEP** to Wangtoo (HPPTCL) 400 kV D/c [Quad^{\$}] line (about 18 kms)
- 2 nos. of 400 kV bays (GIS) at Wangtoo S/s (HPPTCL)

(ii) Final System (To be matching with generation schedule ie.1st July 2026)

- Establishment of 2x315 MVA (7x105 MVA 1-ph units including one spare unit), 400/220 kV GIS Pooling Station at Jhanggi [along with future provision]
- Extension of Wangtoo (HPPTCL) - Shongtong HEP 400 kV (Quad)^{\$} D/c line upto Jhanggi PS with one circuit through Shongtong HEP generation switchyard
- Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line (Twin HTLS)*
- 80 MVAR switchable line reactor at Panchkula end on each circuit of 400 kV Wangtoo (HPPTCL) - Panchkula (PG) D/c line
- 125 MVAR, 420kV Bus reactor at Jhanggi PS (1-ph units along with one spare unit)

\$ Line capacity shall be 2500 MVA per circuit at Nominal voltage

** with minimum capacity of 2100 MVA on each circuit at Nominal voltage*

*** Associated 400 kV bays at Shongtong Generation switchyard shall be under the scope of applicant/generation developer*

##Tidong HEP- Jhanggi PS 220 kV D/C line (along with associated bays at both ends) - under the scope of applicant/generation developer.

Matter regarding implementation of 220kV bays at Jhanggi PS under ISTS for Tidong HEP DTL shall be referred to NCT as GNA Transition process is yet to be completed and BG requirement shall also need to be fulfilled by M/s STATKRAFT in this regard.

2.0 Modifications in “Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5GW) (Jaisalmer/Barmer Complex)”

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5GW) (Jaisalmer/Barmer Complex) was discussed and agreed in 14th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 23.12.22 and 61st NRPC meeting held on 26.12.22. In the 12th NCT meeting held on 24.03.23, above scheme was discussed. In the meeting, various aspects like system strength (SCR), requirement of Barmer-I PS as well as high angular separation in the proposed 765 kV Jalore-Mandsaur D/C (length 320 kms) inter-regional transmission line was discussed.

Grid-India mentioned that in case both lines trip, the angular difference would further increase to 37 degrees. Therefore, in future, in case both lines trip and need to be revived during peak solar generation time, the system may not be stable. New intermediate substation in between may also be proposed and the transmission line length may be reduced as switching of 320 km long inter-regional transmission line may lead to issues in future. In the meeting CTUIL mentioned that 'N-1-1' is a rare contingency and this too may not occur simultaneously during peak solar generation hours. CTUIL also informed regarding possibility of LILO of the above line in future which will reduce the line length and an additional corridor is being planned from Jalore S/s. So, in that condition, angular control will be much better in future.

Based on various above aspects, it was decided that CTUIL would review the transmission scheme in consultation with CEA and GRID-India and the scheme would be discussed in the next NCT meeting.

Subsequently, various joint study meetings were held for review and phasing of transmission scheme among CEA, CTUIL and GRID-INDIA **(copy of minutes enclosed in Annexure-II)**. List of above meetings are as under:

- **Joint study meeting among CEA, CTUIL and GRID-INDIA on 17.04.23 under the Chairmanship of Member (PS), CEA**
- **Joint study meeting among CEA, CTUIL and GRID-INDIA on 18.04.23 & 19.04.23**

Based on deliberations in NCT meeting, system studies were again carried out and discussed in above joint study meetings.

- **Joint study meeting among CEA, CTUIL and GRID-INDIA on 17.04.23 under the Chairmanship of Member (PS), CEA**

In the meeting, CTUIL further stated that different methodologies for calculation of SCR were discussed in the meeting held on 11.04.23 convened by CEA. In the meeting, it was also decided that further deliberation is required to decide on SCR calculation methodology in Indian context. Meanwhile, CTUIL has calculated SCR based on conventional approach for SCR. Further, as discussed in the meeting, in case of hybrid RE contracted capacity i.e. power likely to be injected is being used in SCR calculation and SCR has been observed to be above 5 on Fatehgarh-IV (Section-2) and Barmer-I PS (assuming 50% capacity is evacuated at 220 kV & remaining 50% at 400 kV level pooling stations).

CTUIL stated that the transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 2: 7.5 GW) (Jaisalmer/Barmer Complex) may be implemented in phases. From the studies, it emerged that in first phase, 3 GW RE potential at Fatehgarh-IV (Section-2) (wind: 2 GW, Solar: 1 GW) may be evacuated with net evacuation capacity of 2 GW. Broad transmission scheme for 2 GW RE evacuation at Fatehgarh-IV (Section-2) PS in first phase is as under:

Phase-I

- Establishment of 765/400/220 kV Fatehgarh-IV PS (Section-2)
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line (2nd) at Fatehgarh-IV (Sec-2) PS
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line
- Beawar- Mandsaur PS 765 kV D/c line
- WR scheme (Mandsaur PS & Kurawar PS along with associated system)

In evacuation of RE capacity beyond 2 GW Fatehgarh-IV PS (Section-2) through above corridor, critical loading is observed on 765 kV Beawar -Mandsaur D/c line (N-1 Contingency ≥ 3500 MW) due to huge rush of power towards WR. Therefore, for evacuation of balance 5.5 GW RE power, [Fatehgarh-IV (Section-2) PS: 2 GW and Barmer-I PS: 3.5 GW], following corridors are required:

- Establishment of 765/400/220 kV Barmer-I PS and 765/400 kV Jalore PS
- 765 kV Barmer-I-Jalore and 400 kV Fatehgarh-III -Barmer-I D/c line
- 765 kV Fatehgarh-IV -Jalore -Mandsaur D/c line

In case only one of above corridors is considered, overloading (N-1 contingency) on 765 kV Fatehgarh-III- Fatehgarh-IV line or 400 kV Fatehgarh-III - Barmer-I line is observed. Therefore, both the above corridors are required in same time frame.

In the meeting, Grid-India stated that under N-1 contingency, with outage of one circuit of Jalore - Mandsaur 765 kV D/c line, the angular difference is around 25 degrees. However, in case of tripping of both circuits, the angular difference would increase to 37 degrees which may cause problems in synchronization of lines during peak solar generation hours. The issue of large voltage rise during charging of this long line (>320 km) was also raised by Grid-India. Grid- India proposed termination of this line at an intermediate sub-station so as to reduce the transmission line length which will in turn result in reduction in both angular separation as well as voltage rise during charging of line. Grid - India also suggested to increase load of Rajasthan in winter/summer solar maximized scenario.

In the reply, CTUIL stated that under N-1 contingency, the angular difference is under the stipulated planning criteria. N-1-1 or N-2 is a rare contingency and this too may not occur simultaneously during peak solar hours and deliberation is required on N-1-1/N-2 compliance in planning studies as it may incur additional investment for strengthening of transmission system. Regarding the high voltage rise issue highlighted by Grid-India during line charging, CTUIL stated that the issue of voltage rise is generally dealt by providing adequate reactive compensation with the line/lines as well as corresponding buses. CTU also stated that there are number of 765 kV lines in NR (>320 km) on which voltage rise is within stipulated voltage limits during charging/restoration, as adequate reactive compensation (bus/line) provided on such transmission lines specially the transmission lines planned for RE evacuation.

CTUIL mentioned that as per the TSP practices, in case of split system restoration, 35-degree angle is kept in the synchroscope for line synchronization whereas in normal cases, it is 15 degrees. Grid - India mentioned that as per the recent experiences gained in Western Rajasthan complexes having low short circuit strength and problems faced during synchronization of lines, it will be prudent to plan the system such that parameters are within limits. Grid-India opined that system may be planned in such a way that split system under N-2 contingency doesn't exceed 30 degree angular difference so as to keep some operational margins.

In the meeting CEA opined that Khavda -IV generation (7 GW) along with its associated transmission system may also be considered in study files as both the schemes are required in the same timeframe.

CTUIL stated that as per the inputs provided by SECI, Sirohi and Jalore both have RE potential & are in proximity to each other. Therefore, location of Jalore PS may be shifted towards Sirohi district, so that length of the Jalore - Mandsaur 765 kV D/c line (Original length: 320 kms) is reduced by 30-40 kms. With reduction in line length (new length: 280 kms), the angular difference further reduces and may be considered in final proposal. CEA and Grid-India agreed that this would help the system and Sirohi PS should be preferred over Jalore.

After deliberations, it was decided that revised studies will be carried out considering modification suggested by Grid-India and CEA and revised file will be circulated. Based on outcome of revised file, mitigating measures for N-2 non-compliance of Jalore - Mandsaur 765 kV D/c line (Preferably 30 degree or below) shall be explored.

- **Joint study meeting among CEA, CTUIL and GRID-INDIA on 18.04.23 & 19.04.23**

CTUIL stated that in order to control angular difference in N-1-1/N-2 contingency, various options were considered and it is suggested that in place of Jalore/ Sirohi - Mandsaur 765 kV D/c line, a new substation near Rishabdeo/Salumbar in Udaipur district may be also planned with Sirohi – Rishabdeo/Salumbar 765 kV D/c line (170 km) and Rishabdeo/Salumbar S/s – Mandsaur 765 kV D/c line (160 km) along with LILO of one circuit of 765 kV Chittorgarh - Banaskanta D/c line at Rishabdeo/Salumbar S/s (20 km) for additional anchoring.

CEA opined that initially, Jalore/Sirohi – Mandsaur 765 kV D/c line may be planned for evacuation of power, which may be LILOed at Rishabdeo/Salumbar S/s at later stage as and when N-1-1 non-compliance issue on 765 kV Sirohi- Mandsaur D/c line arises. CTUIL stated that from the studies, it has emerged that LILO would be required if RE injection at Fatehgarh-IV PS (Section-2)/Barmer-I PS exceeds 2.4 GW capacity. CEA suggested that phasing may be explored to defer the investment on above LILO at Rishabdeo/Salumbar.

Grid-India stated that transmission scheme should be planned in such a way that it may evacuate 7.5 GW RE capacity (Fatehgarh-IV: 4 GW, Barmer-I: 3.5 GW) in all possible scenario (winter/summer solar maximized) irrespective of RE type (Wind/Solar) and BESS capacity as the break up RE capacity (solar/wind) is not yet final and it is possible that whole 7.5 GW capacity may comprise of only solar or solar + BESS. This will result in almost 90-100% dispatch in all scenarios.

CTUIL stated that RE potential and dispatches are considered based on report of “Transmission system for integration of over 500GW RE capacity by 2030” as well as SECI/MNRE inputs. As per the above report, 100% solar dispatch and 50% wind dispatch considered in summer solar max scenario, whereas in winter solar max scenario 100% solar dispatch and 10% wind dispatch considered. Details are as under:

Pooling Station	Total RE Potential (by 2030)		RE Potential (by 2027) for which system is planned	Net (Max) Dispatch Considered by 2027(GW) in summer	Net (Max) Dispatch Considered by 2027(GW) in Winter
	Source	Capacity (GW)	Capacity (GW)		
Fatehgarh-IV	Wind	6	4	4	2.4
	Solar	6	4		
	BESS	4	2		
Barmer-I	Wind	3	3	3.5	2.3
	Solar	4	3		
	BESS	1.5	1		
	Total (GW)			7.5	4.7

At present, a comprehensive scheme is planned for Hybrid RE potential (7.5 GW) with 7.5 GW RE dispatch in summer season and 4.7 GW RE dispatch in winter season as Wind contribution reduces significantly in winters. In case the scheme needs to be planned for 7.5 GW RE dispatch in winter scenario also, additional transmissions system is required due to critical loading beyond Jalore/Sirohi.

CEA stated that as the system is being planned for 4 GW injection at Fatehgarh-IV (Section-2), it should cater to 4 GW injection at any instant. CTUIL stated that with change in above approach on RE dispatch levels and considering all solar injection instead of wind/BESS, transmission requirement in RE Planning studies would increase.

CEA stated that with above change in approach, identified transmission scheme shall be able to cater lower quantum of RE potential. Therefore, RE quantum which can be evacuated through revised transmission scheme may be studied. With development of wind or BESS capacity also in the above complex, evacuation capacity may increase from same transmission system.

After conducting studies with incorporation of increased load of Rajasthan as suggested by Grid- India, CTUIL stated that transmission system is adequate for evacuation of about 5.5 GW RE power (solar) in summer & winter scenario. However, with development of wind or BESS capacity in above complex, evacuation capacity shall increase from 5.5 GW. Accordingly, studies were found to be in order for the proposed transmission system.

GRID-India vide mail 20.04.23 informed that the proposed scheme is agreed and the proposal as decided in Joint study meeting meeting on 19.04.23 may be put up to NRPC for further approval. Meanwhile, studies for off-peak scenario may also be carried out and suitable compensation

(if voltage related issues are observed) may be planned later on such that same also gets commissioned in matching time frame of the proposed scheme. Further the scheme was also deliberated & approved in the 65th NRPC meeting held on 21.04.23 subject to deliberations in CMETS-NR meeting.

From the studies it was observed that in Feb solar maximized scenario (revised case) loading of 400kV RAPS-Shujalpur D/c line is marginally higher (about 925 MW) in N-1 contingency. The loading of above line will be reviewed with progress of RE generation projects at Rajasthan and strengthening requirement will be identified later, if required. CTU also carried out the studies for off-peak scenario and same was sent vide mail 27.04.23. From the off-peak studies it was observed that voltages are within limits with proposed reactive compensation requirement. NRLDC informed that they will provide their comments on off peak file in next 3-4 days. It was decided that same shall be suitably included in the minutes.

Subsequently, NRLDC vide mail 03.05.23 sent their observations stating that the bus voltages are observed to be within the limit in the off peak case study based on the compensation given and no major line opening has been considered in order to mitigate high voltage scenario in the off peak case.

Further as advised by Grid-India, Thermal machines (1325 MW) have been switched off in off peak case as these machines are out of service in peak solar case too. NRLDC also stated to correct Qmin limits and machine control mode of few machines in Rajasthan. Based on above, Qmin limit of Ramgarh & Rajwest machines were modified in off peak file by CTU, however, some of machines i.e. Bhadla-3, Bikaner-NW, Mada, Amarsagar etc. are not in service in all scenarios, therefore any change in Qmin limit/machine control mode of these machines will not impact the bus voltages.

With above modifications, studies for off peak were reviewed by CTU. It is observed that there is no significant impact on Grid voltages in revised off peak file and bus voltage are within limits in Rajasthan RE complex with proposed reactive compensation in subject scheme (Rajasthan REZ Ph-IV (Part-2) (Jaisalmer/Barmer Complex))

In view of deliberation held in Joint study meetings as well as in CMETS-NR meeting, following ISTS Transmission scheme was agreed:

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)

Fatehgarh-IV (Section-2) : 4GW and Barmer-I PS: 1.5 GW

Fatehgarh-IV: 4 GW (Considering 2GW at 400kV and 2GW at 220kV)

- Establishment of 4x1500 MVA, 765/400 kV & 5x500 MVA[^], 400/220 kV Fatehgarh- IV (Section-2) Pooling Station along with 2x240 MVA (765 kV) Bus Reactor & 2x125 MVA (400 kV) Bus Reactor*
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) along with 50 MVA switchable line reactor on each ckt at each end (~200 km)
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS along with 330 MVA switchable line reactors at Fatehgarh-IV PS end of each ckt of 765 kV Fatehgarh-IV- Beawar D/c line** (formed after LILO) (~15 km)

- Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAR switchable line reactor on each circuit at each end (~260 km)
- Augmentation by 1x1500 MVA, 765/400 kV ICT at Fatehgarh-II PS (7th)
- 6 nos. of 220 kV line bays at Fatehgarh-IV PS (for RE connectivity)
- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and TBC (2 nos.) at Fatehgarh- IV (Section-2) Pooling Station
- 1 set 400kV Sectionalization bay at Fatehgarh- IV (Section-2) Pooling Station

Future provisions at Fatehgarh-IV PS is already approved in 8th NCT meeting dated 25.03.22

[^]incl 1x500MVA ICT to fulfill 'N-1' requirement

***Incl. spare one ICT unit (500MVA) and spare one reactor unit (80MVAR) at Fatehgarh-IV PS**

****Incl spare one reactor unit (110MVAR) at Fatehgarh-IV PS**

A. Barmer-I : 1.5 GW (Considering 0.5GW at 400kV and 1 GW at 220kV)

- Establishment of 3x1500 MVA, 765/400 kV & 2x500 MVA, 400/220 kV Barmer-I Pooling Station along with 2x240 MVAR (765 kV) Bus Reactor & 2x125 MVAR (400 kV) Bus Reactor*

Future provisions at Barmer-I S/s:

Space for

- 765/400kV ICT along with bays- 3 no.
- 765 kV line bays along with switchable line reactors – 4 nos.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays –4
- 400 kV line bays along with switchable line reactor –4 nos.
- 400/220kV ICT along with bays -8 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalization bays: 2 sets
- 220 kV line bays for connectivity of RE Applications -10 nos.
- 220kV Sectionalization bay: 3 sets
- BC (4 nos.) & TBC (4 nos.)
- STATCOM (2x±300MVAR) along with MSC (4x125 MVAR) & MSR (2x125 MVAR)
- Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad)(~50 km)
- 4 nos. of 220 kV line bays at Barmer-I PS (for RE connectivity)

***Incl. spare one ICT unit (500MVA) and spare one reactor unit (80MVAR) at Barmer-I PS**

B. Common Transmission System for Fatehgarh-IV and Barmer-I in NR

- Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVar (765 kV) & 2x125 MVar (400 kV) Bus Reactor *

Future provisions at Sirohi S/s:

Space for

- 765/400kV ICT along with bays- 4 no.
- 765 kV line bays along with switchable line reactors – 4 no.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 no.
- 400 kV line bays –4 no.
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bay - 6 no.
- 220kV line bays -10 no.
- 220kV Sectionalization bay: 2 sets
- BC (3 nos.) & TBC (3 nos.)
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar)
**Incl. spare one ICT unit (500MVA) and spare one reactor unit (80MVar & 110MVar) at Sirohi PS*

- Establishment of, 765 kV Substation at suitable location near Rishabdeo/Salumbar (Distt. Udaipur) along with 2x240 MVar (765 kV) Bus Reactor*

Future provisions at Rishabdeo/Salumbar S/s:

Space for

- 765/400kV ICT along with bays- 5 no. along with spare unit
- 765 kV line bays along with switchable line reactors – 4 no.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 no.
- 400 kV line bays –4 no.
- 400 kV Bus Reactor along with bay: 3 nos.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bay - 6 nos.
- 220kV line bays -10 no.
- 220kV Sectionalization bay: 2 sets
- BC (3 nos.) & TBC (3 nos.)
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar)

***Incl. spare one reactor unit (80MVar)**

- Fatehgarh-IV (Section-2) PS – Sirohi PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end (~240 km)
- Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end (~200 km)
- Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVar switchable line reactor for each circuit at Sirohi PS end (Quad) (~160 km)
- Sirohi PS- Rishabdeo/Salumbar 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at Sirohi end (~170 km)
- Rishabdeo/Salumbar - Mandsaur PS 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at Rishabdeo/Salumbar end (~160 km)
- LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo/Salumbar S/s (20 km)

In addition to the above, following transmission scheme is also required in Western region for further dispersal of power for above RE complexes (Fatehgarh-IV and Barmer-I).

C. Common Transmission System for Fatehgarh-IV (4GW) and Barmer-I (3.5GW) in Western Region

- Establishment of 765kV Mandsaur Pooling Station along with 2x330MVar (765kV) Bus Reactors (with 1x110MVar & 1x80MVar, 765kV spare single phase reactor unit for line/bus reactor)

Future Scope:

Space for

- 765/400kV ICT along with bays- 6 no.
- 765 kV line bays along with switchable line reactors – 6 nos.
- 765kV Bus Reactor along with bay: 2 no.
- 765kV Sectionalizer bay: 1 -set
- 400 kV line bays along with switchable line reactor – 12 nos.
- 400/220kV ICT along with bays -8 nos.
- 400 kV Bus Reactor along with bay: 2 no.
- 400kV Sectionalization bay: 1- set
- 220 kV line bays: 16 nos.
- 220kV Sectionalization bay: 2 sets
- 220kV BC and TBC: 3 nos.
- STATCOM (± 300 MVar) along with MSC (2x125 MVar) & MSR (1x125 MVar): 1 no.
- Mandsaur PS – Indore(PG) 765 kV D/c Line (200km) along with 1x330MVar switchable line reactor (SLR) on each ckt at Mandsaur end
- Establishment of 765/400 (2x1500MVA) 400/220 (2x500MVA) & 220/132kV (3x200MVA) Kurawar S/s (with 1x500MVA spare single phase transformer unit) with 2x330MVar 765kV bus reactor and 1x125MVar 420kV bus reactor (with 1x110MVar & 1x80MVar, 765kV spare single phase reactor unit for line/bus reactor)

Future Scope:

Space for

- 765/400kV ICT along with bays- 4 no.
- 765 kV line bays along with switchable line reactors – 8 nos.
- 765kV Bus Reactor along with bay: 2 no.
- 765kV Sectionalizer bay: 1 -set
- 400 kV line bays along with switchable line reactor – 10 nos.
- 400/220kV ICT along with bays -6 nos.
- 400 kV Bus Reactor along with bay: 2 no.
- 400kV Sectionalization bay: 1- set
- 220 kV line bays: 12 nos.
- 220kV Sectionalization bay: 2 sets
- 220kV BC and TBC: 3 nos.
- 220/132kV ICT along with bays: 3 nos.
- 132kV line bays: 8
- 132kV Sectionalization bay: 1 set
- STATCOM (± 300 MVar) along with MSC (2x125 MVar) & MSR (1x125 MVar): 1 no.
- Mandsaur – Kurawar 765kV D/c line (~235km.) with 240MVar switchable line reactors at both ends
- LILO of Indore – Bhopal 765kV S/c line at Kurawar (LILO route length ~15km.)
- Kurawar – Ashtha 400kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~65km)
- LILO of one circuit of Indore – Itarsi 400kV D/c line at Astha (LILO route length ~ 30km.)
- Shujalpur – Kurawar 400kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~40km)

Estimated Cost : Rs 14600 Cr (NR portion)

: Rs 5967Cr (WR portion)

Implementation of the transmission scheme may be phased out and Barmer-I PS and associate transmission system may not be required in Phase-1 scheme. However, the evacuation capacity is indicated as per the revised approach for dispatching and considering only solar. The Phase-I transmission scheme is as under :

Phase-I (1.5 GW RE evacuation capacity)

- Establishment of 3x1500 MVA*, 765/400 kV & 2x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station along with 2x240 MVar (765 kV) Bus Reactor & 2x125 MVar (400 kV) Bus Reactor*

- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) along with 50 MVar switchable line reactor on each ckt at each end (~200 km)
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS along with 330 MVar switchable line reactors at Fatehgarh-IV PS end of each ckt of 765 kV Fatehgarh-IV- Beawar D/c line** (formed after LILO) (~15 km)
- Beawar- Mandsaur PS 765 kV D/c line along with 240 MVar switchable line reactor on each circuit at each end (~260 km)
- Establishment of 765kV Mandsaur Pooling Station[§] along with 2x330MVar (765kV) Bus Reactors (with 1x110MVar & 1x80MVar, 765kV spare single phase reactor unit for line/bus reactor)
- Mandsaur PS – Indore(PG) 765 kV D/c Line (200km) along with 1x330MVar switchable line reactor (SLR) on each ckt at Mandsaur end
- Establishment of 765/400 (2x1500MVA) 400/220 (2x500MVA) & 220/132kV (3x200MVA) Kurawar S/s[§] (with 1x500MVA spare single phase transformer unit) with 2x330MVar 765kV bus reactor and 1x125MVar 420kV bus reactor (with 1x110MVar & 1x80MVar, 765kV spare single phase reactor unit for line/bus reactor)
- Mandsaur – Kurawar 765kV D/c line (~235km.) with 240MVar switchable line reactors at both ends
- LILO of Indore – Bhopal 765kV 765kV S/c line at Kurawar (LILO route length ~15km.)
- Kurawar – Ashtha 400kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~65km)
- LILO of one circuit of Indore – Itarsi 400kV D/c line at Astha (LILO route length ~ 30km.)
- Shujalpur – Kurawar 400kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~40km)

**Incl. spare one ICT unit (500MVA) and spare one reactor unit (80MVar) at Fatehgarh-IV PS*

***Incl spare one reactor unit (110MVar) at Fatehgarh-IV PS*

§along with future scope provision as per original scheme

Further, in order to evacuate additional 4 GW RE (Fatehgarh-IV (Section-2) PS: 2.5 GW and Barmer-I PS: 1.5 GW), comprehensive transmission system including above Phase-I scheme is mentioned as above.

Annexure-X

List of Participants of 18th Consultation meeting for Evolving Transmission Schemes in NR held on 28.04.2023

CEA

Shri Nitin Deswal	Assistant Director
Shri Kanhaiya	Assistant Director

SECI

Shri R.K.Agarwal	Consultant
Shri Prashant	

Grid India

Shri Gaurav Singh	Manager
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CTU

Shri Jasbir Singh	CGM (CTU)
Shri Kashish Bhambhani	GM (CTU)
Shri Sandeep Kumawat	Ch. Manager (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri Tej Prakash Verma	Ch. Manager (CTU)
Shri R Narendra Sathvik	Manager (CTU)
Shri Madhusudan Meena	Engineer (CTU)

DTL

Shri Dinesh Singh	Manager(T) SS&LM. Plg.
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RVVNL

Shri Surendra Vashistha	Superintending Engineer
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HPPTCL

Shri Rajneesh Kumar GM (C&D)

Shri Harmanjeet Singh

HPPCL

Shri Deshraj GM

Shri Arun Goyel Advisor HPPCL Shimla

Shri M. K. Chopra DGM

LTA/Connectivity Applicants

Shri Lalit Mogha Cannice Renewables Energy Private Limited

Shri Pavan Kumar Gupta Juniper Green Beta Private Limited

Shri Pritpal Singh JSW Neo Renew Energy Five Limited

Shri Pritesh Lodha Proteus Energy Private Limited

Shri H R Shah Rajasthan Solarpark Development Company Limited

Shri Kapil Verma Rajasthan Solarpark Development Company Limited

Shri Hariom Gupta Rajasthan Solarpark Development Company Limited

Shri Arzaan Dordi Serentica Renewables India Private Limited

Shri Sandeep Anand Tepsol Saur Urja Private Limited

Smt Vishalakshi Malan THDC India Limited

Shri R.M.Dubey THDC India Limited

Shri Rakesh Singh THDC India Limited

Shri Mallikarjuan Udayagiri THDC India Limited

Shri Sanjeev Kumar Tata Power Trading Company Limited

Shri Himanshu Puri Tidong Power Generation Private Limited

Shri Jaspreet Latawa	Tidong Power Generation Private Limited
Shri Susheel Kumar	Tidong Power Generation Private Limited
Shri Anurag Sharma	Tidong Power Generation Private Limited

I/27306/2023

Annexure-I



भारत सरकार
Government of India

विद्युत मंत्रालय
Ministry of Power

केन्द्रीय विद्युत प्राधिकरण
Central Electricity Authority

विद्युत प्रणाली योजना एवं मूल्यांकन - I प्रभाग
Power System Planning & Appraisal - I Division

सेवा में / To

1. COO, CTUIL, Saudamini Plot 2, Sector 29, Gurugram-122001
2. The General Manager, HPPTCL, Himfed Bhawan, Panjari, Shimla-171005
3. The Director (Electrical), HPPCL Himfed Building, BCS, New Shimla-171009
4. The Manager, Statkraft, Unit 401, 4th Floor, Salcon Rasvilas, Saket, New Delhi-110017

विषय / Subject: Meeting regarding the transmission system for evacuation of power from
Shongtong Karcham HEP (STKHEP) in Himachal Pradesh

महोदय / Sir,

The minutes of the meeting held through VC on 12.04.2023 regarding the transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) in Himachal Pradesh is attached herewith.

संलग्न / Encl: यथोपरि / as above

भवदीय / Yours faithfully,

(Signature)
18/04/2023

(नितिन देसवाल / Nitin Deswal)

उप निदेशक /Deputy Director

प्रति लिपि / CC: SA to Member (PS), CEA

I/27306/2023

Minutes of the meeting held on 12.04.2023 regarding the transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) in Himachal Pradesh

A meeting had been convened on 12.04.2023 regarding the transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) in Himachal Pradesh.

List of participants is attached as Annexure.

Background:

The transmission scheme 'Transmission system for evacuation of power from Shongtong Karcham HEP (450 MW) and Tidong HEP (150 MW)' was agreed in the 11th meeting of National Committee on Transmission (NCT) held on 28.12.2023 & 17.01.2023 with following broad scope of works to be implemented in 2 phases:

Phase-I (with Tidong HEP and with schedule: 01st July 2026)

- Establishment of 2x315 MVA, 400/220 kV GIS Pooling Station at Jhangi
- Jhangi PS – Wangtoo (Quad) 400 kV D/c line
- 2 nos. of 400 kV bays at Wangtoo (HPPTCL) for termination of 400 kV Jhangi PS – Wangtoo D/c line

Phase-II (with Shongtong HEP and with schedule: 31st July 2026]

- LILO of one circuit of Jhangi PS – Wangtoo (HPPTCL) 400 kV D/c (Quad) line at generation switchyard of Shongtong HEP
- Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line
- 400 kV bays at Wangtoo S/s (2 nos.) and Panchkula S/s (2 nos.) for termination of Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line

Note: Tidong HEP- Jhangi PS 220 kV D/C line - under the scope of applicant/generation developer

The transmission scheme had been recommended by the NCT to MoP for implementation through TBCB route.

Subsequently, HPPCL vide their letter dated 05.04.2023 intimated that the commissioning date of Shongtong Karcham HEP (STKHEP) had been revised from July 2026 to July 2025 and requested to review the transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) in Himachal Pradesh due to the preponement in the commissioning timeline of STKHEP.

Deliberations in the meeting:

HPPCL informed that in the recent deliberations between Govt. of HP and management of HPPCL, progress of the STKHEP was reviewed and the various issues/disputes that were delaying the project were resolved. Further, the civil contractor has committed to complete the project by July 2025 by increasing the manpower, machinery and other resources. In view

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of this, HPPCL has revised the commissioning schedule to July 2025 and hence, there would be need for transmission system for evacuation of power from STKHEP by July 2025.

M/s Statkraft (for Tidong HEP) stated that for Tidong HEP, the requirement of Jhangi PS would be from July 2026, which was mentioned in their Connectivity application. Statkraft added that since the location of Jhangi Pooling station has not yet been finalized and considering the issue of tough terrain and local RoW issues, constructing dedicated line from Tidong HEP to Jhangi Pooling station would take considerable time and hence, the requirement of transmission system for Tidong HEP would be from July 2026.

CEA stated that in the case of different timelines for the requirement of transmission system for Tidong HEP and STKHEP, the transmission system could be phased accordingly.

CTUIL opined that the time frame for implementation of the transmission system by July 2025 does not seem feasible; hence, the minimum time in which the transmission system for evacuation of power from STKHEP could be implemented, needs to be determined considering the hilly terrain. CTUIL added that the downstream system i.e. Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line would also be required for evacuation of power from STKHEP.

On a query about the topography of the area, CTUIL (Engg wing) informed that these are very high altitude areas with long periods of no activity in winter and monsoon season. CTUIL opined that considering the various difficulties in construction of transmission line in that area, the completion of the transmission system in July 2025 timeframe does not seem to be practical.

CTUIL stated that in the meeting held on 25.11.2022 wherein implementation schedule of transmission lines was deliberated and it was agreed that the schedule of transmission system to be implemented in challenging areas like hilly terrain may be taken as 30/36 months. CEA agreed and stated that the minimum timeframe for hilly terrain is to be considered as 30-36 months.

CTUIL suggested that the earlier proposed interim arrangement of LILO of one ckt. of Baspa – Wangtoo 400 kV D/c line at Shongtong HEP (with LILO length portion of about 6 kms) could also be explored by HPPCL as an alternative till the commissioning of the final transmission system.

HPPCL stated that from the total length of 54 kms of the 400 kV transmission line from Jhangi PS to Wangtoo (HPPTCL), the 18 km portion from Wangtoo (HPPTCL) to Shongtong HEP could be implemented by July 2025, which would be sufficient for evacuation of power from STKHEP till the completion of complete final transmission system.

CTUIL stated that the Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line is part of the LTA system for STKHEP and would be required for evacuation of power; however, as an interim arrangement, the 18 km section could be considered but sufficient time period should be provided to the TSP for completion of that portion.

HPPCL requested that the small portion of 18 km of the transmission line should be implemented by July 2025 to evacuate power from STKHEP.

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On query from CEA about the bifurcation of the transmission scheme in two phases, CTUIL stated that the 18 km portion of the line could be considered as an interim arrangement; however, the power transfer from STKHEP could not be committed and would be on real time basis depending on the available margins in the system. STKHEP may have to apply for T-GNA for evacuation of power, till the Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line is implemented.

Further, CTUIL added that in case the interim transmission system becomes ready for commissioning and the generation (STKHEP) get delayed, then the generator has to pay the charges as per the prevailing regulation.

On the implementation time for construction of transmission line in that area, M/s Statkraft stated that the considering the local issues and no activity period of 3 months during winters, the 24 months' time period is very slim time line. HPPTCL stated that considering the tough terrain, lot of efforts would be required to complete the 18 km portion of transmission line in 24 months; however, for evacuation of power from STKHEP the option could be explored.

After further deliberation following was agreed:

1. The transmission system to be bifurcated in two parts as mentioned below:
 - (i) Interim Part (with time frame of 24 months)
 - Generation switchyard of Shongtong HEP to Wangtoo (HPPTCL) 400 kV D/c line (about 18 kms)
 - 2 nos. of 400 kV bays at Wangtoo (HPPTCL)
 - (ii) Final System (with time frame of 1st July 2026)
 - Establishment of 2x315 MVA (7x105 MVA 1-ph units including one spare unit), 400/220 kV GIS Pooling Station at Jhangi
 - Completion of remaining portion of 400 kV line i.e. from Shongtong HEP to Jhangi PS so as to form Jhangi PS – Wangtoo (Quad) D/c line with LILO of one circuit of Jhangi PS – Wangtoo (HPPTCL) 400 kV D/c (Quad) line at generation switchyard of Shongtong HEP
 - Wangtoo (HPPTCL) - Panchkula (PG) 400 kV D/c line
 - 400 kV bays at Wangtoo S/s (2 nos.) and Panchkula S/s (2 nos.) for termination of 400 kV Wangtoo (HPPTCL) - Panchkula (PG) D/c line
2. To expedite the implementation of the transmission scheme, HPPCL/HPPTCL would facilitate the TSP implementing the scheme in obtaining various clearances.
3. In the interim period till the commissioning of the final system, HPPCL would have to apply for T-GNA for evacuation of power from STKHEP.
4. The matter would be taken up for discussion in the next meeting of NCT.

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Annexure**List of participants:****• CEA**

- 1) Shri Ishan Sharan (Chief Engineer)
- 2) Smt Manjari Chaturvedi (Director)
- 3) Shri Nitin Deswal (Dy. Director)
- 4) Ms Komal Dupare (Asst. Director)

• CTUIL

- 1) Shri Sourov Chakraborty (Sr. GM Engg.)
- 2) Shri Kashish Bhambhani (GM Plg)

• HPPTCL

- 1) Shri Rajeev Sood (Director P&C)
- 2) Shri Rajneesh Kumar (GM C&D)
- 3) Shri Harmanjeet Singh (Dy. Manager)

• HPPCL

- 1) Shri Desh Raj (GM-Electrical)
- 2) Shri Arun Goyal (Advisor)

• Statkraft (for Tidong HEP)

- 1) Shri Himanshu Puri
- 2) Shri Anurag Sharma

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Annexure-II



भारत सरकार

Government of India

विद्युत मंत्रालय

Ministry of Power

केन्द्रीय विद्युत प्राधिकरण

Central Electricity Authority

विद्युत प्रणाली योजना एवं मूल्यांकन-I प्रभाग

Power System Planning & Appraisal-I Division

सेवा में / To,

1. COO (CTUIL), Saudamini, Plot no. 2, Sector -29, Gurgaon-122 001
2. Director (System Operation), Grid-India, B-9, Qutab Institutional Area, Katwaria Sarai, New Delhi- 110010

विषय /Subject: Minutes of the meeting to review the "Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5GW) (Jaisalmer/Barmer Complex)"

महोदय/ Sir,

Please find enclosed the minutes of the meeting held on 17.04.2023, 18.04.2023 & 19.04.2023 to review the "Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5 GW) (Jaisalmer/Barmer Complex)".

भवदीय / Yours faithfully,

(मंजरी चतुर्वेदी/Manjari Chaturvedi)

निदेशक/Director

21/4/2023

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Minutes of the meeting held on 17.04.2023, 18.04.2023 & 19.04.2023 to review the “Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 2: 7.5 GW) (Jaisalmer/Barmer Complex)”

List of participants is enclosed as Appendix-I.

A. Background:

- (i) In the 12th NCT meeting held on 24.03.2023, transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2: 7.5 GW) (Jaisalmer/ Barmer Complex) was deliberated, wherein, various aspects like system strength (SCR), requirement of Barmer-I PS as well as high angular separation in the proposed Jalore - Mandsaur 765 kV D/c line (length 320 kms) inter-regional transmission line were discussed. Grid-India also highlighted the large number the trippings that have taken place in the Rajasthan RE complex during last one year.
- (ii) After detailed deliberations, it was decided that CTUIL would review the transmission scheme in consultation with CEA and Grid-India and the scheme would be discussed in the next NCT meeting.

B. Deliberations in the joint study meeting held on 17.04.2023 amongst CEA, CTUIL and Grid - India under the Chairmanship of Member (PS), CEA:

- (i) CTUIL stated that the transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 2: 7.5 GW) (Jaisalmer/Barmer Complex) may be implemented in phases. From the studies, it emerged that in first phase, 3 GW RE potential at Fatehgarh-IV (Section-2) (wind: 2 GW, Solar: 1 GW) may be evacuated with net evacuation capacity of 2 GW.

Broad transmission scheme for 2 GW RE evacuation at Fatehgarh-IV (Section-2) PS in first phase is as under:

Phase-I

- Establishment of 765/400/220 kV Fatehgarh-IV PS (Section-2)
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line (2nd) at Fatehgarh-IV (Sec-2) PS
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line
- Beawar- Mandsaur PS 765 kV D/c line
- WR scheme (Mandsaur PS & Kurawar PS along with associated system)

In evacuation of RE capacity beyond 2 GW Fatehgarh-IV PS (Section-2) through above corridor, critical loading is observed on 765 kV Beawar -Mandsaur D/c line (N-1 Contingency ≥ 3500 MW) due to huge rush of power towards WR. Therefore, for evacuation of balance 5.5 GW RE power, [Fatehgarh-IV (Section-2) PS: 2 GW and Barmer-I PS: 3.5 GW], following corridors are required:

- Establishment of 765/400/220 kV Barmer-I PS and 765/400 kV Jalore PS

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- 765 kV Barmer-I-Jalore and 400 kV Fatehgarh-III -Barmer-I D/c line
- 765 kV Fatehgarh-IV -Jalore -Mandsaur D/c line

In case only one of the above corridors is considered, overloading (N-1 contingency) on 765 kV Fatehgarh-III- Fatehgarh-IV line or 400 kV Fatehgarh-III - Barmer-I line is observed. Therefore, both the above corridors are required in same time frame. Details of scheme is as under.

Phase-II: 5.5 GW RE evacuation [Fatehgarh-IV (Section-2) PS (2 GW) and Barmer-I PS (3.5 GW)]

- Establishment of 765/400/220 kV Barmer-I PS and 765/400 kV Jalore PS
 - Fatehgarh-IV (Section-2) PS – Jalore 765 kV D/c line
 - Fatehgarh-III (Sec-2) PS – Barmer-I 400 kV D/c line
 - Barmer-I – Jalore 765 kV D/c line
 - Jalore- Mandsaur PS 765 kV D/c line
- (ii) CTUIL further stated that at present, 2.6 GW of stage-II connectivity has already been received at Fatehgarh-IV PS (Section-2). However, M/s Adani also submitted their dissent for GNA transition for their 1.5 GW solar project in Fatehgarh-III PS as per CERC (Connectivity & GNA to the ISTS) Regulation 2022. With above developments and GNA effectiveness from 05.04.2023, deliberations are required for phasing of implementation of the scheme proposed as above.
- (iii) CTUIL further stated that different methodologies for calculation of SCR were discussed in the meeting held on 11.04.2023 convened by CEA. In the meeting, it was also decided that further deliberations are required to decide on SCR calculation methodology in Indian context. Meanwhile, CTUIL has calculated SCR based on conventional approach for SCR calculation. Further, as discussed in the meeting, in case of hybrid RE contracted capacity i.e. power likely to be injected is being used in SCR calculation and SCR has been observed to be above 5 on Fatehgarh-IV (Section-2) and Barmer-I PS (assuming 50% capacity is evacuated at 220 kV & remaining 50% at 400 kV level pooling stations).
- (iv) Grid-India stated that under N-1 contingency, with outage of one circuit of Jalore - Mandsaur 765 kV D/c line, the angular difference is around 25 degrees. However, in case of tripping of both circuits, the angular difference would increase to 37 degrees which may cause problems in synchronization of lines during peak solar generation hours. The issue of large voltage rise during charging of this long line (>320 km) was also raised by Grid-India. Grid-India proposed termination of this line at an intermediate sub-station so as to reduce the transmission line length which will in turn result in reduction in both angular separation as well as voltage rise during charging of line. Grid - India also suggested to increase load of Rajasthan in winter/summer solar maximized scenario.
- (v) CTUIL stated that under N-1 contingency, the angular difference is under the stipulated planning criteria. N-1-1 or N-2 is a rare contingency and this too may not occur simultaneously during peak solar hours and deliberations are required on N-1-1/

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N-2 compliance in planning studies as it may incur additional investment for strengthening of transmission system. Regarding the “high voltage rise” issue highlighted by Grid-India during line charging, CTUIL stated that the issue of voltage rise is generally dealt by providing adequate reactive compensation with the line/lines as well as corresponding buses. CTU also stated that there are number of 765 kV lines in NR (>320 km) on which voltage rise is within stipulated voltage limits during charging/restoration, as adequate reactive compensation (bus/line) provided on such transmission lines especially the transmission lines planned for RE evacuation.

- (vi) Grid-India mentioned that as per Transmission Planning Criteria, 2023:

“The lines for which the angular difference between its terminal buses is more than 20 degree after contingency of one circuit may be selected for performing stability studies.”

However, as angular separation is around 25 degrees and transient stability studies are not carried out by CTUIL, it is proposed to plan the system such that angular separation is reduced to less than 20 degrees under N-1 contingency.

- (vii) CTUIL mentioned that as per the Transmission Service Provider (TSP) practices, in case of split system restoration, 35-degree angle is kept in the synchroscope for line synchronization whereas in normal cases, it is 15 degrees.
- (viii) Grid - India mentioned that as per the recent experiences gained in Western Rajasthan complexes having low short circuit strength and problems faced during synchronization of lines, it will be prudent to plan the system such that parameters are within limits. Grid-India opined that system may be planned in such a way that split system under N-2 contingency doesn't exceed 30 degree angular difference so as to keep some operational margins.
- (ix) CEA opined that Khavda -IV generation (7 GW) along with its associated transmission system may also be considered in study files as both the schemes are required in the same timeframe. This may improve the loadings of NR-WR corridors i.e. 400 kV RAPP-Shujalpur and 765 kV Jalore -Mandsaur D/c line.
- (x) CTUIL stated that as per the inputs provided by SECI, Sirohi and Jalore both have RE potential and are in proximity to each other. Therefore, location of Jalore PS may be shifted towards Sirohi district, so that length of the Jalore - Mandsaur 765 kV D/c line (Original length: 320 kms) is reduced by 30-40 kms. With reduction in line length (new length: 280 kms), the angular difference further reduces and may be considered in final proposal. CEA and Grid-India agreed that this would help the system and Sirohi PS should be preferred over Jalore.
- (xi) After deliberations, it was decided that revised studies will be carried out considering modification suggested by Grid-India and CEA and revised file will be circulated. Based on outcome of revised file, mitigating measures for N-2 non-compliance of Jalore - Mandsaur 765 kV D/c line (Preferably 30 degree or below) shall be explored.

C. Deliberations in the joint study meeting held on 18.04.2023 & 19.04.2023 amongst CEA, CTUIL and Grid - India:

- (i) CTUIL stated that in order to control angular difference in N-1-1/N-2 contingency conditions, various options were considered and it is suggested that in place of Jalore/ Sirohi - Mandsaur 765 kV D/c line, a new substation near Rishabdeo/Salumbar in

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- Udaipur district may be also planned with Sirohi – Rishabdeo/Salumbar 765 kV D/c line (170 km) and Rishabdeo/Salumbar S/s – Mandsaur 765 kV D/c line (160 km) along with LILO of one circuit of 765 kV Chittorgarh - Banaskanta D/c line at Rishabdeo/Salumbar S/s (20 km) for additional anchoring.
- (ii) CEA opined that initially, Jalore/Sirohi – Mandsaur 765 kV D/c line may be planned for evacuation of power, which may be LILOed at Rishabdeo/Salumbar S/s at later stage as and when N-1-1 non-compliance issue on 765 kV Sirohi-Mandsaur D/c line arises.
 - (iii) CTUIL stated that from the studies, it has emerged that LILO would be required if RE injection at Fatehgarh-IV PS (Section-2)/Barmer-I PS exceeds 2.4 GW capacity.
 - (iv) CEA suggested that phasing may be explored to defer the investment on above LILO at Rishabdeo/Salumbar.
 - (v) Grid-India stated that transmission scheme should be planned in such a way that it may evacuate 7.5 GW RE capacity (Fatehgarh-IV: 4 GW, Barmer-I: 3.5 GW) in all possible scenario (winter/summer solar maximized) irrespective of RE type (Wind/Solar) and BESS capacity as the breakup of RE capacities (solar/wind) is not yet final and it is possible that whole 7.5 GW capacity may comprise of only solar or solar + BESS. This will result in almost 90-100% dispatch in all scenarios.
 - (vi) CTUIL stated that RE potential and dispatches are considered based on report of **“Transmission system for integration of over 500 GW RE capacity by 2030”** as well as SECI/MNRE inputs.

As per the above report, 100% solar dispatch and 50% wind dispatch considered in summer solar maximum scenario, whereas in winter solar maximum scenario 100% solar dispatch and 10% wind dispatch considered. Details are as under:

Pooling Station	Total RE Potential (by 2030)		RE Potential (by 2027) for which system is planned	Net (Max) Dispatch Considered by 2027(GW) in summer	Net (Max) Dispatch Considered by 2027(GW) in Winter
	Source	Capacity (GW)	Capacity (GW)		
Fatehgarh-IV	Wind	6	4	4	2.4
	Solar	6	4		
	BESS	4	2		
Barmer-I	Wind	3	3	3.5	2.3
	Solar	4	3		
	BESS	1.5	1		
			Total (GW)	7.5	4.7

At present, comprehensive scheme is planned for Hybrid RE potential (7.5 GW) with 7.5 GW RE dispatch in summer season and 4.7 GW RE dispatch in winter season as Wind contribution reduces significantly in winters. In case the scheme needs to be planned for 7.5 GW RE dispatch in winter scenario also, additional transmissions system is required due to critical loading beyond Jalore/Sirohi.

- (vii) CEA stated that as the system is being planned for 4 GW injection at Fatehgarh-IV (Section-2), it should cater to 4 GW injection at any instant under normal operating conditions.

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- (viii) CTUIL stated that with change in above approach on RE dispatch levels and considering all solar injection instead of wind/BESS, transmission requirement in RE Planning studies would increase.
- (ix) CEA stated that with above change in approach, identified transmission scheme shall be able to cater lower quantum of RE potential. Therefore, RE quantum which can be evacuated through revised transmission scheme may be studied. With development of wind or BESS capacity also in the above complex, evacuation capacity may increase from same transmission system.
- (x) After conducting studies with incorporation of increased load of Rajasthan as suggested by Grid- India, CTUIL stated that transmission system is adequate for evacuation of about 5.5 GW RE power (solar) in summer and winter scenario. However, with development of wind or BESS capacity in above complex, evacuation capacity shall increase from 5.5 GW. Accordingly, studies were found to be in order for the transmission system given at Annexure-1 for 5.5 GW capacity. Result of system studies are also enclosed at Exhibit-1.
- (xi) Grid-India stated that the angular separation is as under in the revised scheme:

Scenario	Angular Difference Between Jalore/Sirohi & Rishabdeo in February Solar maximum scenario– Scenario 7 Case	Angular Difference Between Jalore/Sirohi & Rishabdeo in June Solar maximum scenario– Scenario 4 Case
Base Case	9.4°	8.3°
N-1 of 765 kV Jalore – Rishabdeo D/C line	14.5°	12.6°
N-2 of 765 kV Jalore – Rishabdeo D/C line	29.3°	25.6°

Grid- India opined that studies for off-peak scenario may also be carried out and suitable compensation (if voltage related issues are observed) may be planned later on, such that same also gets commissioned in matching time frame of the proposed scheme. The same was agreed.

- (xii) After deliberations, following was agreed in above meetings:
- Comprehensive transmission scheme agreed for evacuation of 5.5 GW RE (Solar) capacity is attached as Annexure-1. In case, wind generation/ RE with BESS gets connected at Fatehgarh-IV/Barmer-I Complex, the same system will be sufficient to evacuate over 5.5 GW Capacity. Major changes in original scheme vis-vis the revised scheme is attached as Annexure-2.
 - Implementation of the transmission scheme may be phased out and Barmer-I PS and associated transmission system may not be required in Phase-I scheme. However, the evacuation capacity is indicated as per the revised approach for dispatching and considering only solar. The broad elements of Phase-I scheme comprises the following:

Phase-I (1.5 GW RE evacuation capacity)

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- Establishment of 765/400/220 kV Fatehgarh-IV PS (Section-2)
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line (2nd) at Fatehgarh-IV (Sec-2) PS
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line
- Beawar- Mandsaur PS 765 kV D/c line
- WR scheme (Mandsaur PS & Kurawar PS along with associated system)

Further, in order to evacuate additional 4 GW RE (Fatehgarh-IV (Section-2) PS: 2.5 GW and Barmer-I PS: 1.5 GW), comprehensive transmission system including above Phase-I scheme is required as per **Annexure-1**.

Annexure-I

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)

Fatehgarh-IV (Section-2): 4 GW and Barmer-I PS: 1.5 GW

- A. Fatehgarh-IV: 4 GW (Considering 2 GW injection at 400 kV and 2 GW injection at 220 kV level)**

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- Establishment of 4x1500 MVA, 765/400 kV & 5x500 MVA[^], 400/220 kV Fatehgarh- IV (Section-2) Pooling Station along with 2x240 MVA_r (765 kV) Bus Reactor & 2x125 MVA_r (400 kV) Bus Reactor*
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) along with 50 MVA_r switchable line reactor on each ckt at each end (~200 km)
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS along with 330 MVA_r switchable line reactors at Fatehgarh-IV PS end of each ckt of 765 kV Fatehgarh-IV- Beawar D/c line** (formed after LILO) (~15 km)
- Beawar- Mandsaur PS 765 kV D/c line along with 240 MVA_r switchable line reactor on each circuit at each end (~260 km)
- 6 Nos. of 220 kV line bays at Fatehgarh-IV PS (for RE connectivity)
- 220 kV Sectionalization bay (1 set) along with BC (2 Nos.) and TBC (2 Nos.) at Fatehgarh- IV (Section-2) Pooling Station
- Augmentation by 1x1500 MVA, 765/400 kV ICT at Fatehgarh-II PS (7th)
- 1 set 400 kV Sectionalization bay at Fatehgarh- IV (Section-2) Pooling Station

Future provisions at Fatehgarh-IV PS is already approved in 8th NCT meeting dated 25.03.22

^including 1x500MVA ICT to fulfill 'N-1' requirement

****Including spare one ICT unit (500MVA) and spare one reactor unit (80MVA_r) at Fatehgarh-IV PS***

*****Including spare one reactor unit (110MVA_r) at Fatehgarh-IV PS***

B. Barmer-I: 1.5 GW (Considering 0.5 GW injection at 400 kV and 1 GW injection at 220 kV level)

- Establishment of 3x1500 MVA, 765/400 kV & 2x500 MVA, 400/220 kV Barmer-I Pooling Station along with 2x240 MVA_r (765 kV) Bus Reactor and 2x125 MVA_r (400 kV) Bus Reactor*

Future provisions at Barmer-I S/s:

Space for

- 765/400 kV ICT along with bays- 3 no.
- 765 kV line bays along with switchable line reactors – 4 nos.
- 765 kV Bus Reactor along with bay: 1 no.
- 400 kV line bays –4 Nos.
- 400 kV line bays along with switchable line reactor –4 nos.
- 400/220 kV ICT along with bays -8 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400 kV Sectionalization bays: 2 sets
- 220 kV line bays for connectivity of RE Applications -10 nos.
- 220 kV Sectionalization bay: 3 sets
- BC (4 nos.) and TBC (4 nos.)
- STATCOM (2x±300MVA_r) along with MSC (4x125 MVA_r) & MSR (2x125 MVA_r)

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- Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad)(~50 km)
- 4 nos. of 220 kV line bays at Barmer-I PS (for RE connectivity)

****Including spare one ICT unit (500 MVA) and spare one reactor unit (80 MVar) at Barmer-I PS.***

C. Common Transmission System for Fatehgarh-IV and Barmer-I in NR

- Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Sirohi along with 2x240 MVar (765 kV) and 2x125 MVar (400 kV) Bus Reactor *

Future provisions at Sirohi S/s:

Space for

- 765/400kV ICT along with bays- 4 no.
- 765 kV line bays along with switchable line reactors – 4 nos.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 nos.
- 400 kV line bays –4 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400 kV Sectionalization bay: 2 sets
- 400/220 kV ICT along with bay - 6 nos.
- 220 kV line bays -10 nos.
- 220kV Sectionalization bay: 2 sets
- BC (3 nos.) and TBC (3 nos.)
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) and MSR (2x125 MVar)

****Including spare one ICT unit (500 MVA) and spare one reactor unit (80 MVar and 110 MVar) at Sirohi PS.***

- Establishment of 765 kV switching station at suitable location near Rishabdeo/Salumbar (Distt. Udaipur) along with 2x240 MVar (765 kV) Bus Reactor *

Future provisions at Rishabdeo/Salumbar S/s:

Space for

- 765/400 kV ICT along with bays- 5 nos. along with spare unit
- 765 kV line bays along with switchable line reactors – 4 nos.
- 765 kV Bus Reactor along with bay: 1 no.
- 400 kV line bays along with switchable line reactor –4 nos.
- 400 kV line bays –4 nos.
- 400 kV Bus Reactor along with bay: 3 nos.
- 400 kV Sectionalization bay: 2 sets
- 400/220kV ICT alongwith bay - 6 nos.
- 220 kV line bays -10 nos.
- 220 kV Sectionalization bay: 2 sets

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- BC (3 nos.) and TBC (3 nos.)
- STATCOM (2x±300MVA_r) along with MSC (4x125 MVA_r) and MSR (2x125 MVA_r)

****Including spare one reactor unit (80 MVA_r)***

- Fatehgarh-IV (Section-2) PS – Sirohi PS 765 kV D/c line along with 240 MVA_r switchable line reactor for each circuit at each end (~240 km)
- Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVA_r switchable line reactor for each circuit at each end (~200 km)
- Sirohi PS-Chittorgarh (PG) 400 kV D/c line along with 80 MVA_r switchable line reactor for each circuit at Sirohi PS end (Quad) (~160 km)
- Sirohi PS- Rishabdeo/Salumbar 765 kV D/c line along with 330 MVA_r switchable line reactor for each circuit at Sirohi end (~170 km)
- Rishabdeo/Salumbar - Mandsaur PS 765 kV D/c line along with 330 MVA_r switchable line reactor for each circuit at Rishabdeo/Salumbar end (~160 km)
- LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo/Salumbar S/s (20 km)

In addition to the above, following transmission scheme is also required in Western region for further dispersal of power for above RE complexes (Fatehgarh-IV and Barmer-I)

D. Common Transmission System for Fatehgarh-IV (4 GW) and Barmer-I (1.5 GW) in Western Region

- Establishment of 765 kV Mandsaur Pooling Station along with 2x330 MVA_r (765 kV) Bus Reactors (with 1x110 MVA_r and 1x80 MVA_r, 765 kV spare single phase reactor unit for line/bus reactor)

Future Provisions: Space for

- 765/400 kV ICT along with bays- 6 nos.
- 765 kV line bays along with switchable line reactors – 6 nos.
- 765 kV Bus Reactor along with bay: 2 nos.
- 765 kV Sectionalizer bay: 1 - set
- 400 kV line bays along with switchable line reactor – 12 nos.
- 400/220 kV ICT along with bays -8 nos.
- 400 kV Bus Reactor along with bay: 2 nos.
- 400 kV Sectionalization bay: 1- set
- 220 kV line bays: 16 nos.
- 220 kV Sectionalization bay: 2 sets
- 220 kV BC and TBC: 3 nos.
- STATCOM (±300MVA_r) along with MSC (2x125 MVA_r) and MSR (1x125 MVA_r): 1 no.
- Mandsaur PS – Indore (PG) 765 kV D/c line (200 km) along with 1x330 MVA_r switchable line reactor (SLR) on each ckt at Mandsaur end

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- Establishment of 765/400 kV (2x1500 MVA) 400/220 kV (2x500 MVA) and 220/132 kV (3x200 MVA) Kurawar S/s (with 1x500 MVA spare single phase transformer unit) with 2x330 MVar, 765 kV bus reactor and 1x125 MVar, 420 kV bus reactor (with 1x110 MVar and 1x80 MVar, 765 kV spare single phase reactor unit for line/bus reactor)

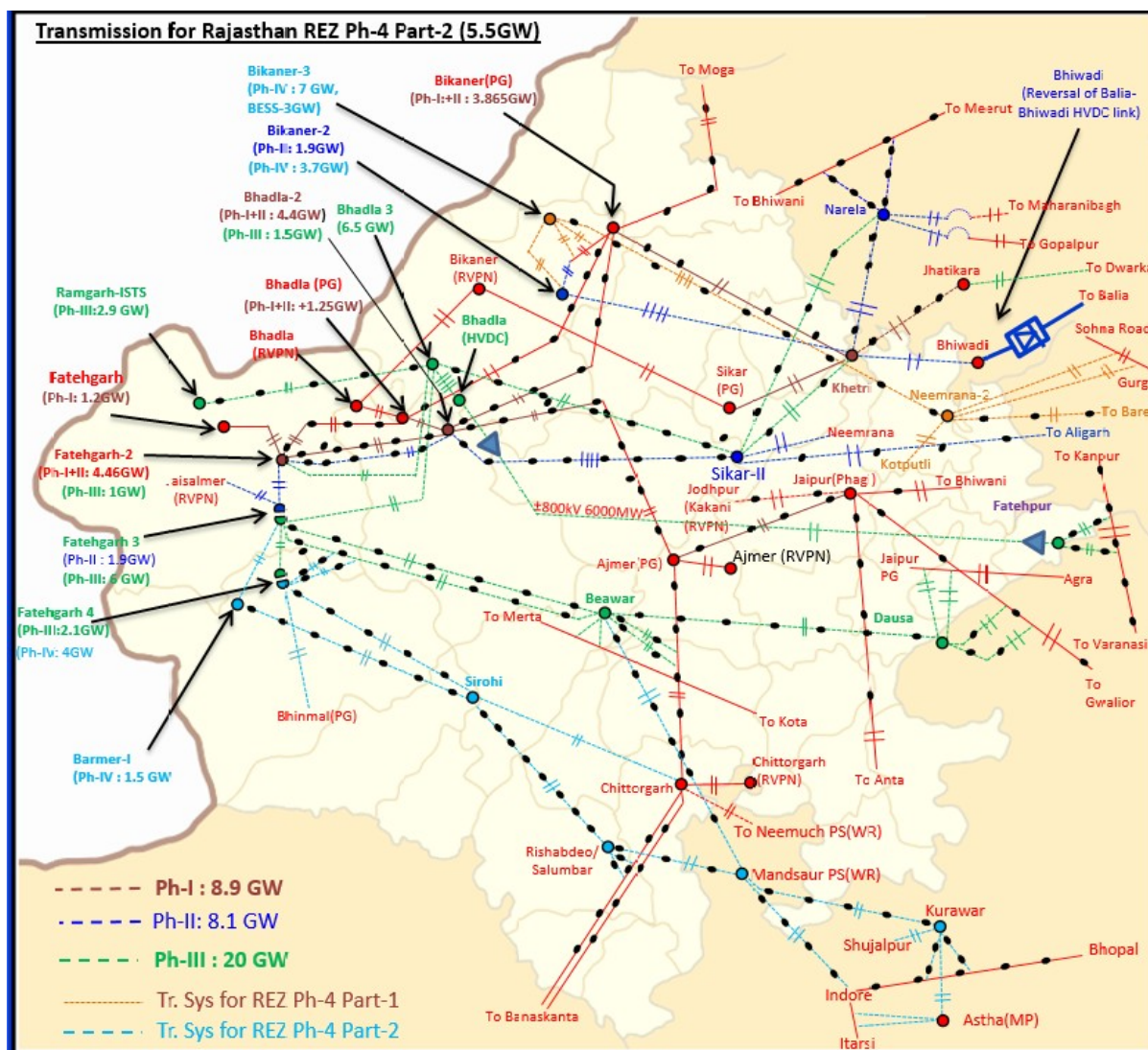
Future Provisions: Space for

- 765/400 kV ICT along with bays- 4 nos.
 - 765 kV line bays along with switchable line reactors – 8 nos.
 - 765 kV Bus Reactor along with bay: 2 nos.
 - 765 kV Sectionalizer bay: 1 - set
 - 400 kV line bays along with switchable line reactor – 10 nos.
 - 400/220 kV ICT along with bays -6 nos.
 - 400 kV Bus Reactor along with bay: 2 nos.
 - 400 kV Sectionalization bay: 1- set
 - 220 kV line bays: 12 nos.
 - 220 kV Sectionalization bay: 2 sets
 - 220 kV BC and TBC: 3 nos.
 - 220/132 kV ICT along with bays: 3 nos.
 - 132 kV line bays: 8 nos.
 - 132 kV Sectionalization bay: 1 set
 - STATCOM (± 300 MVar) along with MSC (2x125 MVar) & MSR (1x125 MVar): 1 no.
- Mandsaur – Kurawar 765 kV D/c line (~235 km) with 240 MVar switchable line reactors at both ends
 - LILO of Indore – Bhopal 765 kV S/c line at Kurawar (LILO route length ~15 km)
 - Kurawar – Ashtha 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~65 km)
 - LILO of one circuit of Indore – Itarsi 400 kV D/c line at Astha (LILO route length ~ 30 km)
 - Shujalpur – Kurawar 400 kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~40 km)

Estimated Cost (NR) : Rs 14556 Cr (NR portion)

: Rs 5967 Cr (WR portion)

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Annexure-II

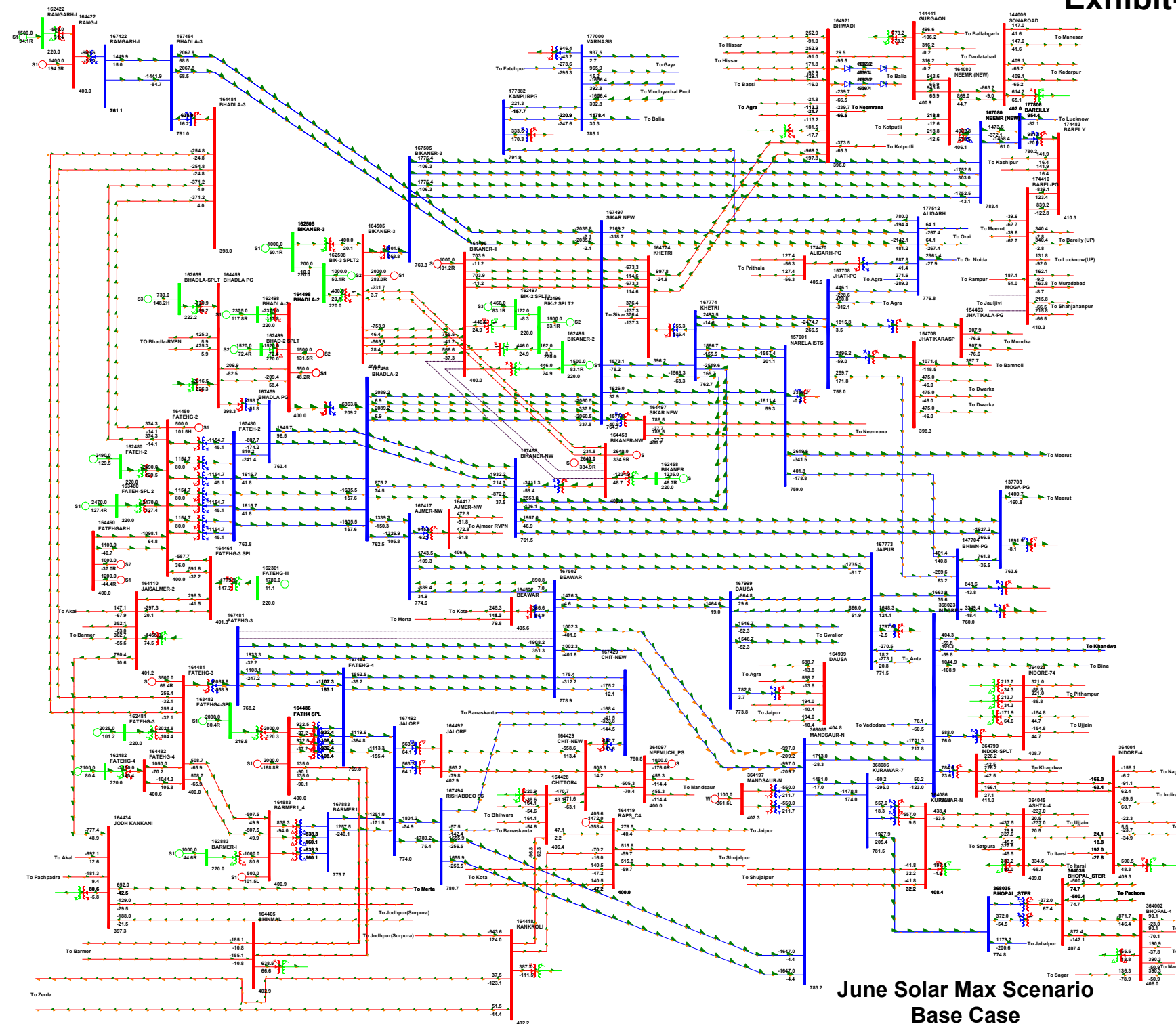
S.No	Original Scope	Revised Scope
1	- Establishment of 4x1500 MVA, 765/400 kV & 5x500 MVA, 400/220 kV Barmer-I Pooling Station along with 2x240 MVA (765 kV) Bus Reactor & 2x125 MVA (400 kV) Bus Reactor* 6 nos. of 220 kV line bays at Barmer-I PS (for RE connectivity)	- Establishment of 3x1500 MVA, 765/400 kV & 2x500 MVA, 400/220 kV Barmer-I Pooling Station along with 2x240 MVA (765 kV) Bus Reactor & 2x125 MVA (400 kV) Bus Reactor* 4 nos. of 220 kV line bays at Barmer-I PS (for RE connectivity)
2	Fatehgarh-IV (Section-2) PS – Jalore PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end (~200 km)	Fatehgarh-IV (Section-2) PS – Sirohi PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end (~240 km)
3	Barmer-I PS– Jalore PS 765 kV D/c line along with 330 MVA switchable line reactor for each circuit at Jalore PS end (~165 km)	Barmer-I PS– Sirohi PS 765 kV D/c line along with 240 MVA switchable line reactor for each circuit at each end (~200 km)
4	Jalore PS -Chittorgarh (PG) 400 kV D/c line along with 50 MVA switchable line reactor for each circuit at each end (Quad) (~200 km)	Sirohi PS -Chittorgarh (PG) 400 kV D/c line along with 80 MVA switchable line reactor for each circuit at Sirohi end (Quad) (~160 km)
5		Establishment of, 765 kV Substation at suitable location near Rishabdeo/Salumbar (Distt. Udaipur) along with 2x240 MVA (765 kV) Bus Reactor
6		LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo/Salumbar S/s (20 km)
7	Jalore PS- Mandsaur PS 765 kV D/c line along with 330 MVA switchable line reactor for each circuit at each end (~320 km)	Sirohi PS- Rishabdeo/Salumbar 765 kV D/c line along with 330 MVA switchable line reactor for each circuit at Sirohi end (~170 km)
8		Rishabdeo/Salumbar - Mandsaur PS 765 kV D/c line along with 330 MVA switchable line reactor for each circuit at Rishabdeo/Salumbar end (~160 km)
7		LILO of one circuit of 765 kV Chittorgarh-Banaskanta D/c line at Rishabdeo/Salumbar S/s (20 km)

Additional investment from earlier scope: Rs 900 Crore

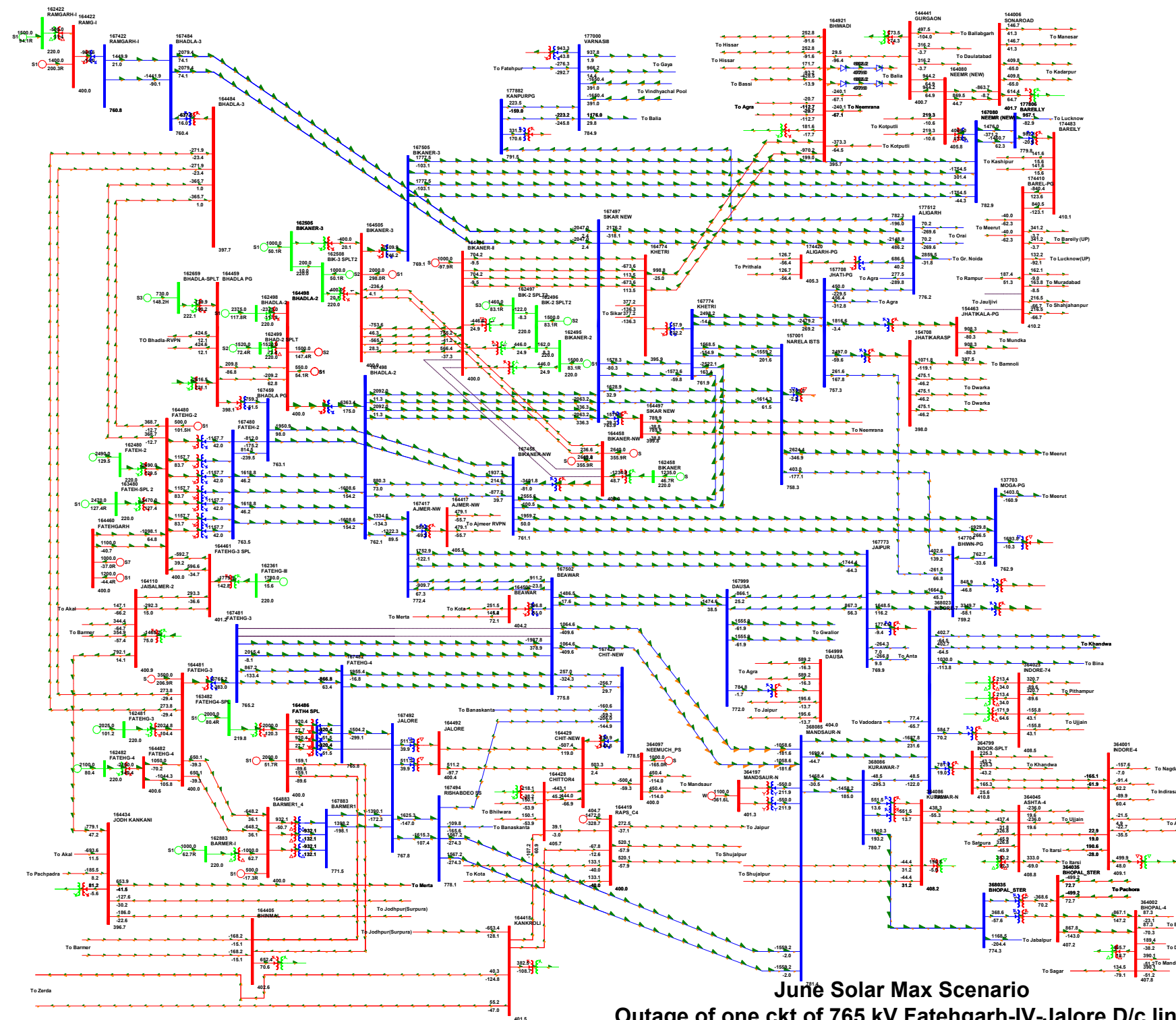
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Appendix I**List of participants:**

Sl. No.	Name (Smt/ Ms/ Shri)	Designation
CEA		
1	A.K. Rajput	Member (Power Systems)
2	Ishan Sharan	Chief Engineer
3	Manjari Chaturvedi	Director
4	Nitin Deswal	Deputy Director
5	Komal Dupare	Asst. Director
6	Kanhaiya Singh Kushwaha	Asst. Director
CTUIL		
7	Kashish Bhambhani	GM
8	Sandeep Kumawat	DGM
9	Narendra Sathvik	Manager
Grid- India		
10	Surajit Banerjee	CGM (NLDC)
11	Vivek Pandey	GM (NLDC)
12	Alok Kumar	GM (NRLDC)
13	Priyam Jain	Manager (NLDC)
14	Gaurav Singh	Manager (NRLDC)
15	Gaurab Das	Asst. Manager (NLDC)
16	Asif Khan	Asst. Manager (NRLDC)

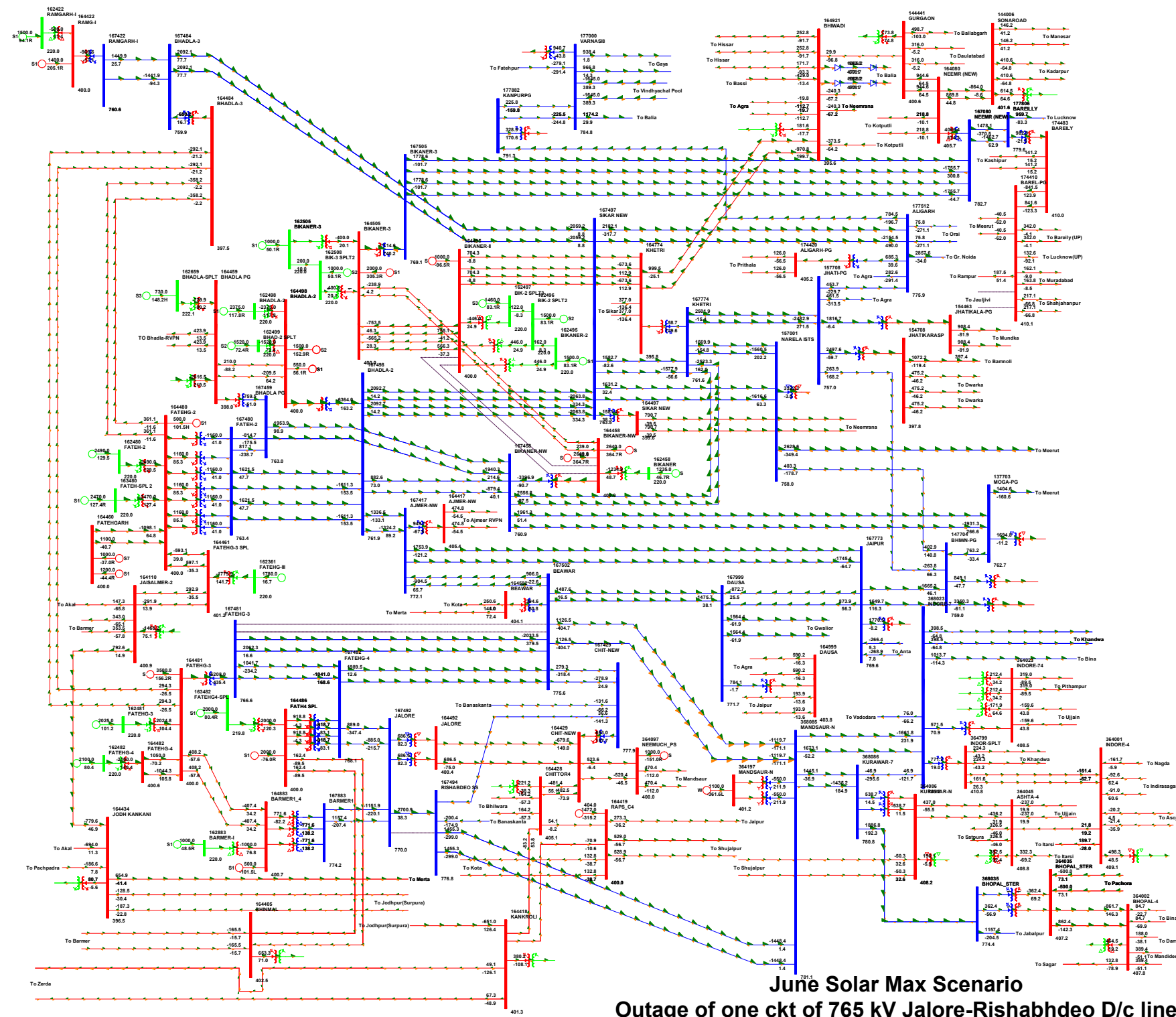


Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)

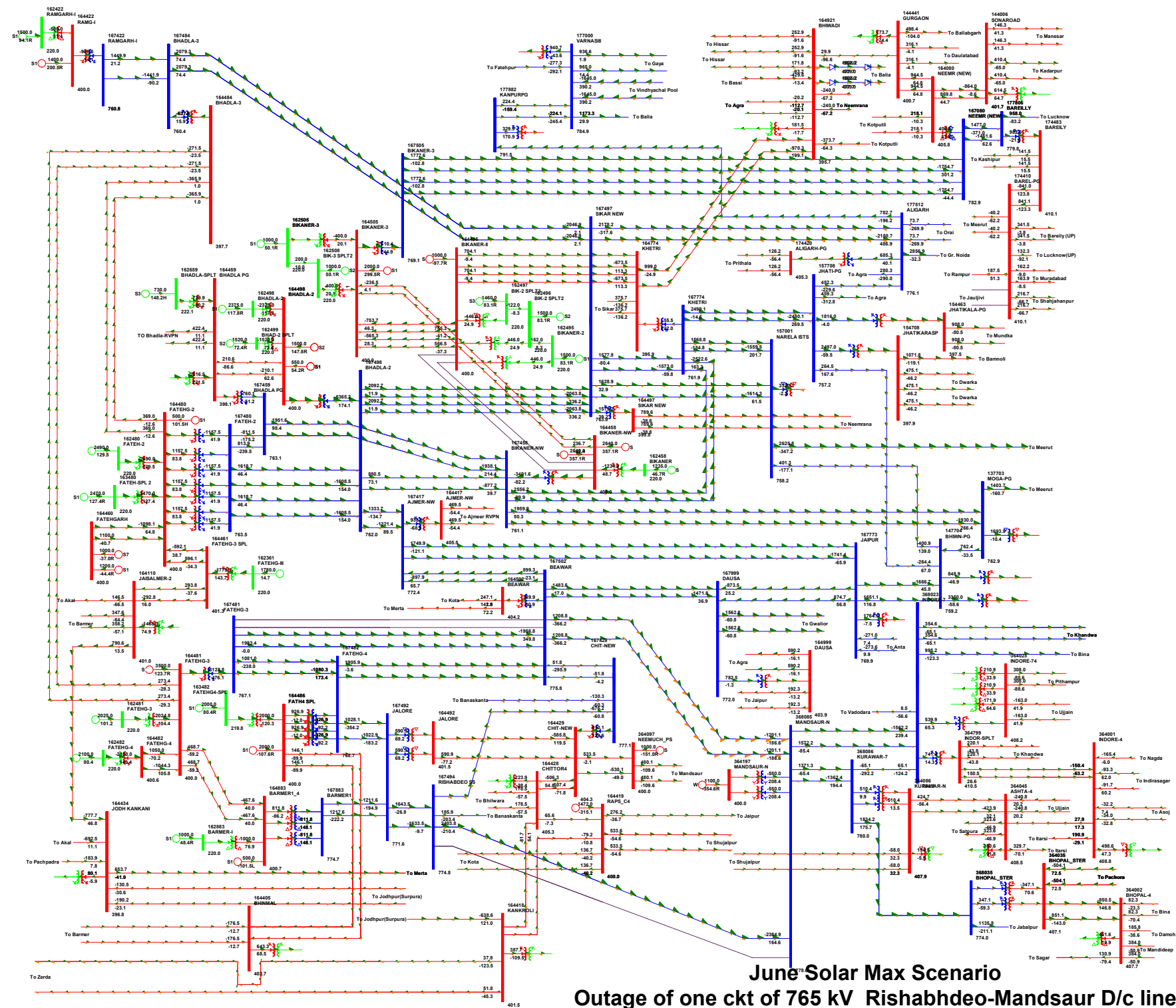


June Solar Max Scenario
Outage of one ckt of 765 kV Fatehgarh-IV-Jalore D/c line

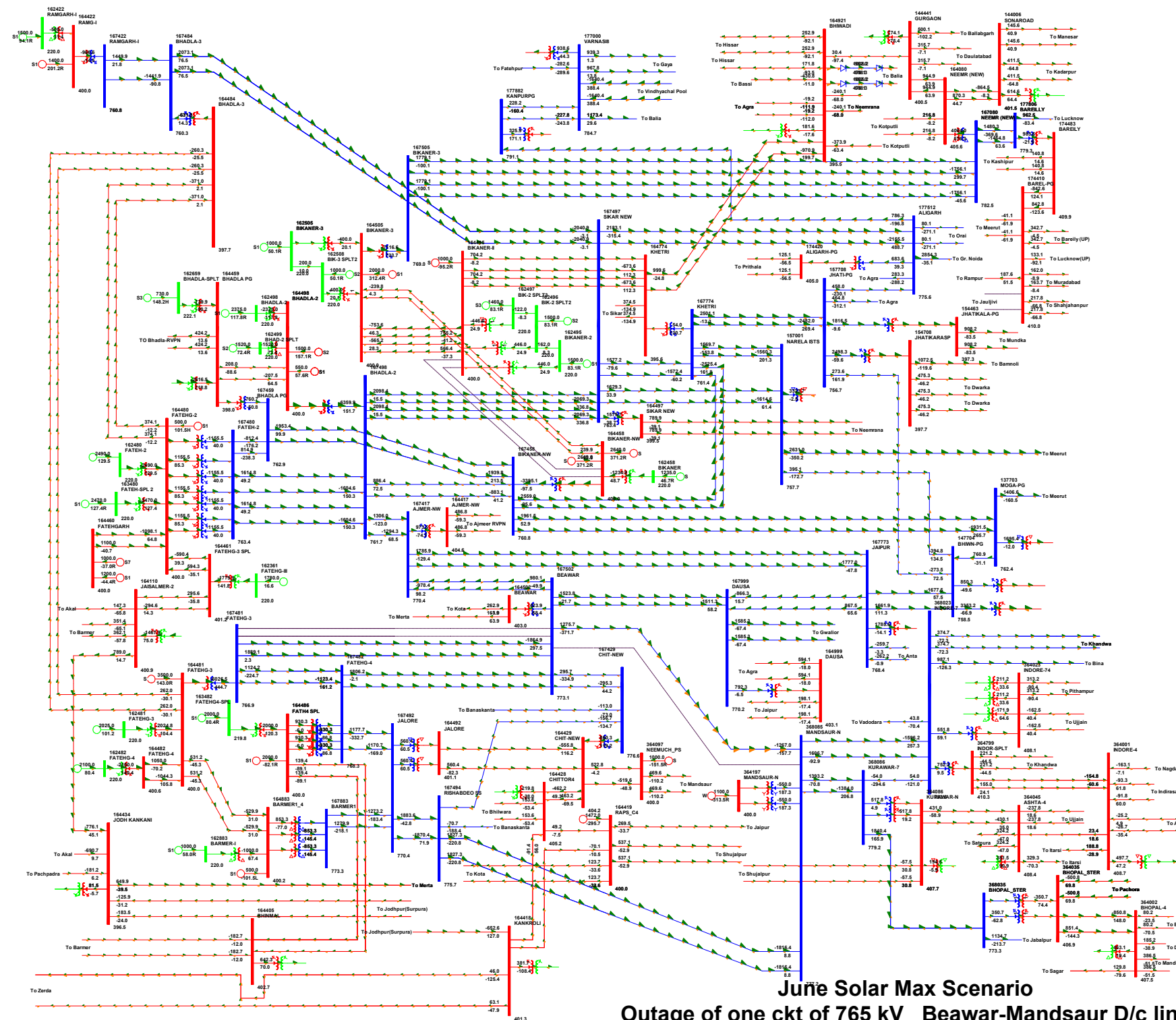
Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



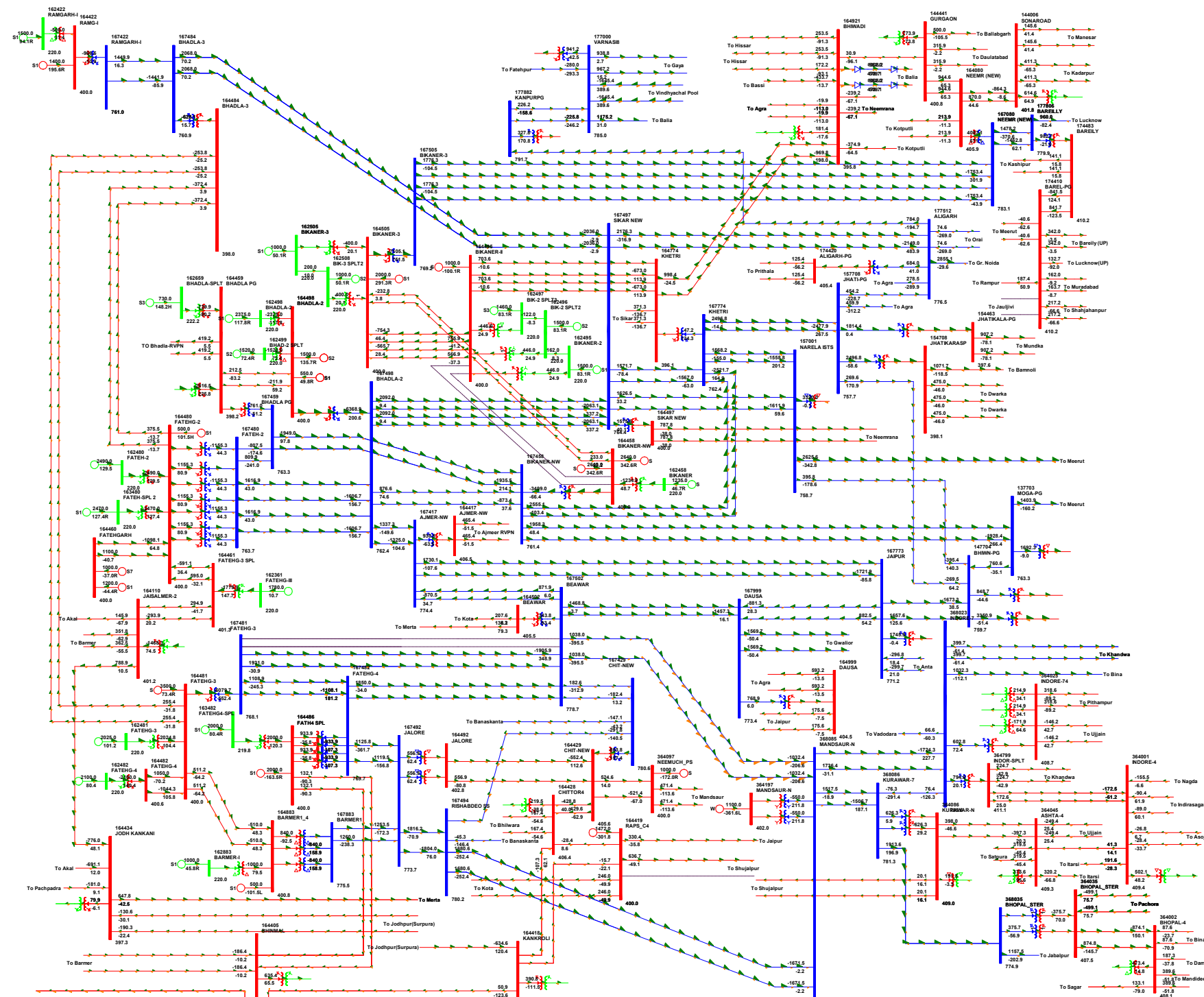
Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



June Solar Max Scenario

Outage of one ckt of 765 kV Beawar-Mandsaur D/c line

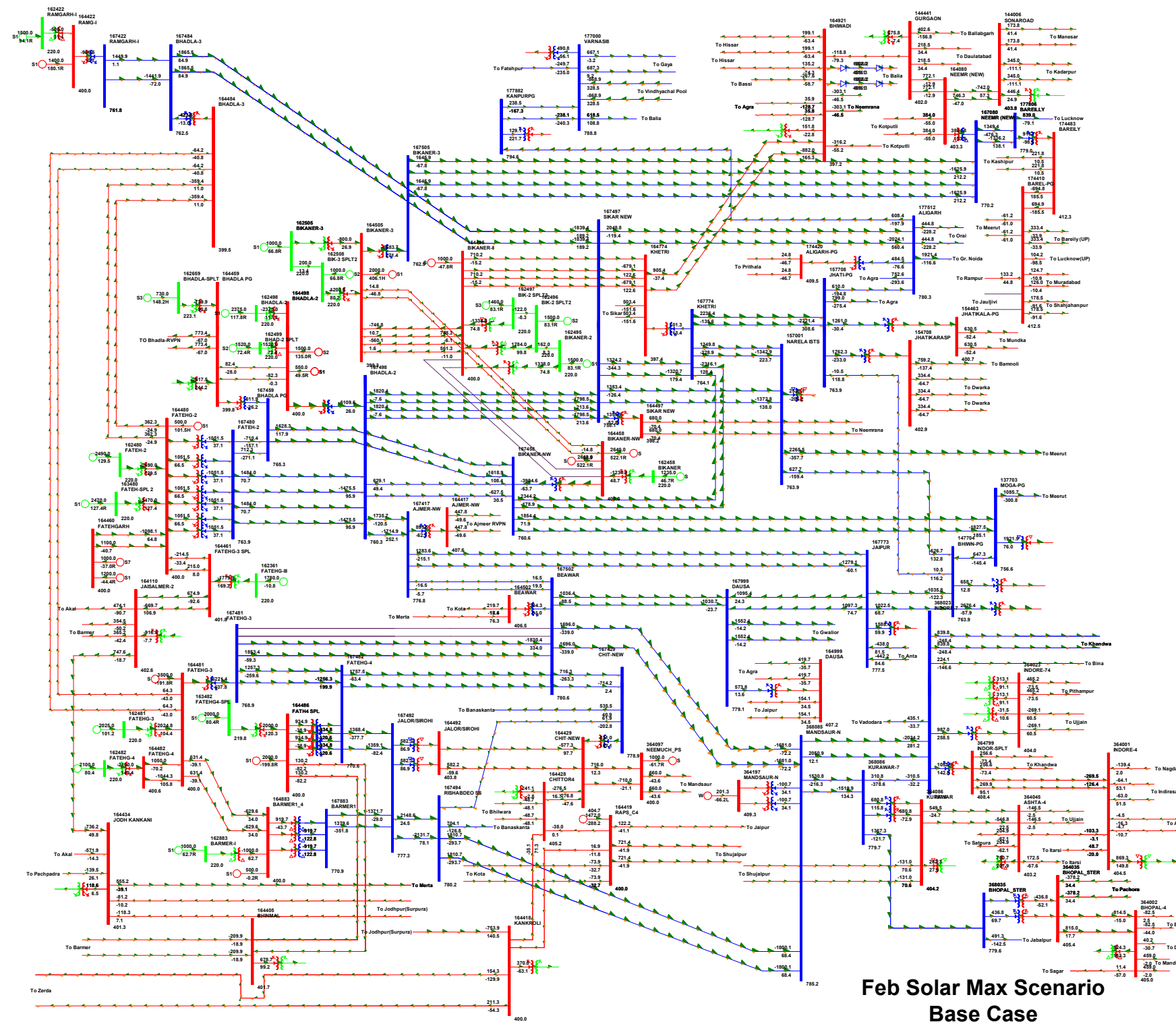
Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



June Solar Max Scenario

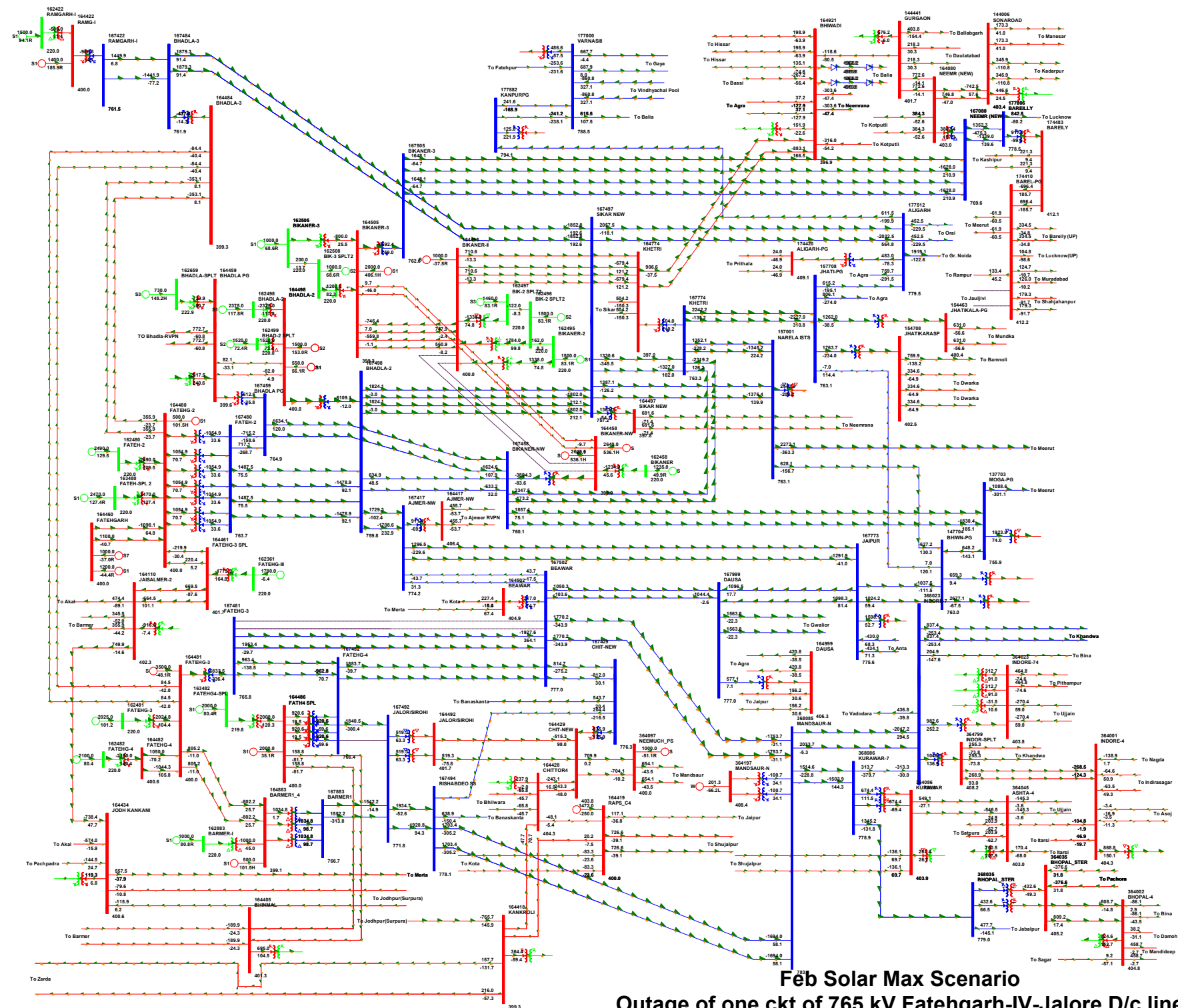
Outage of one ckt of 400 kV RAPS-Sujalpur D/c line

Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)

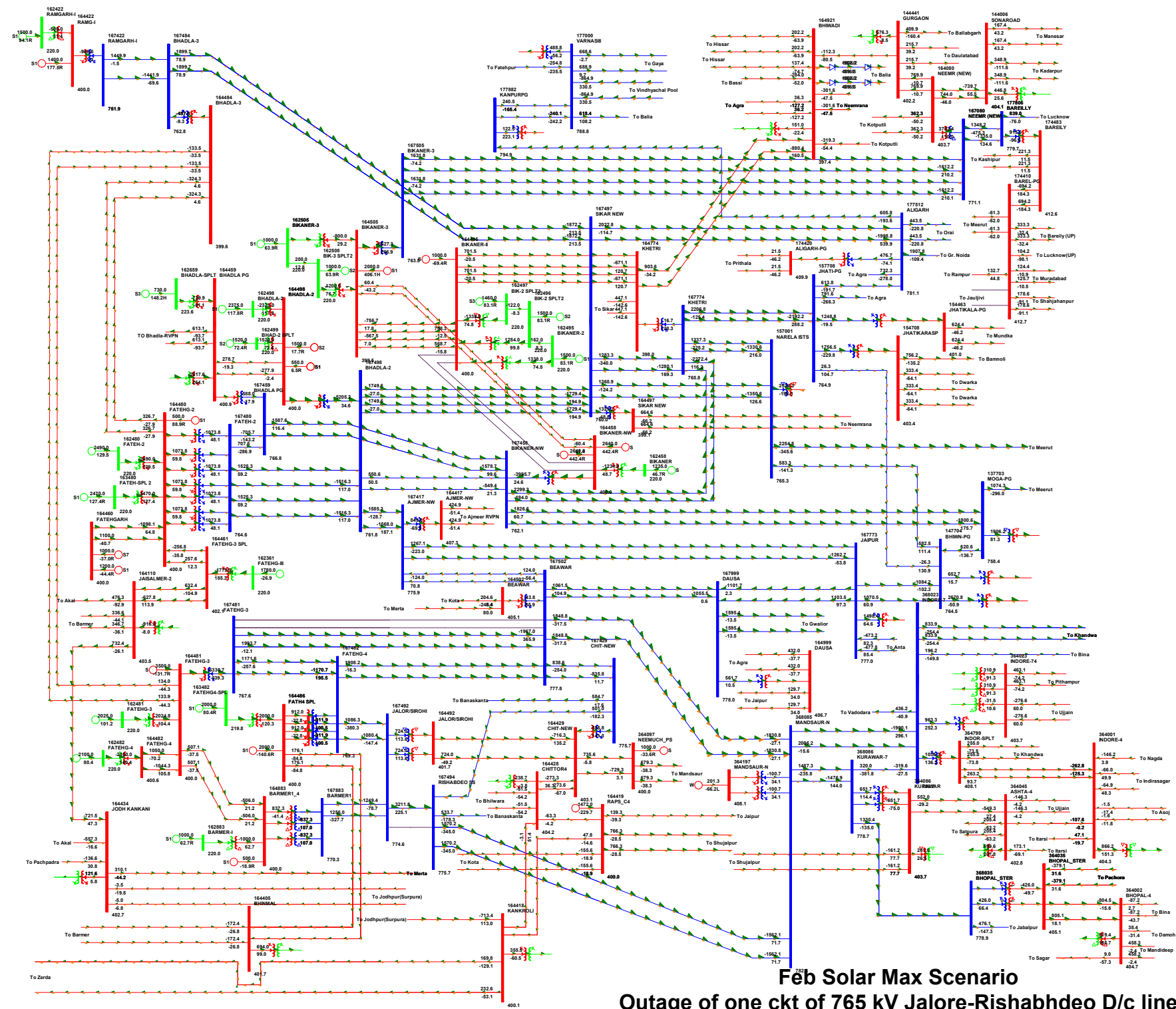


Feb Solar Max Scenario Base Case

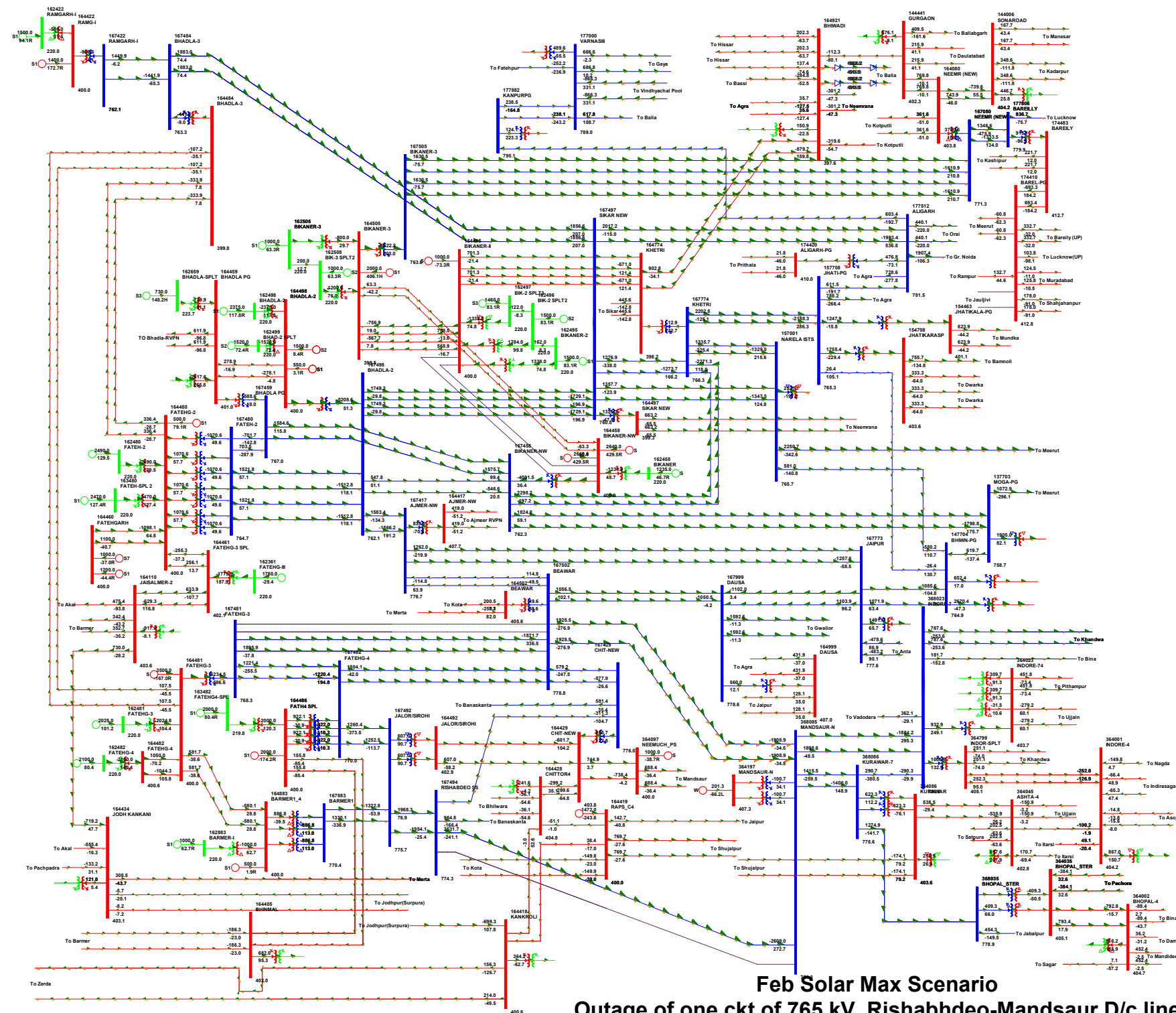
Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



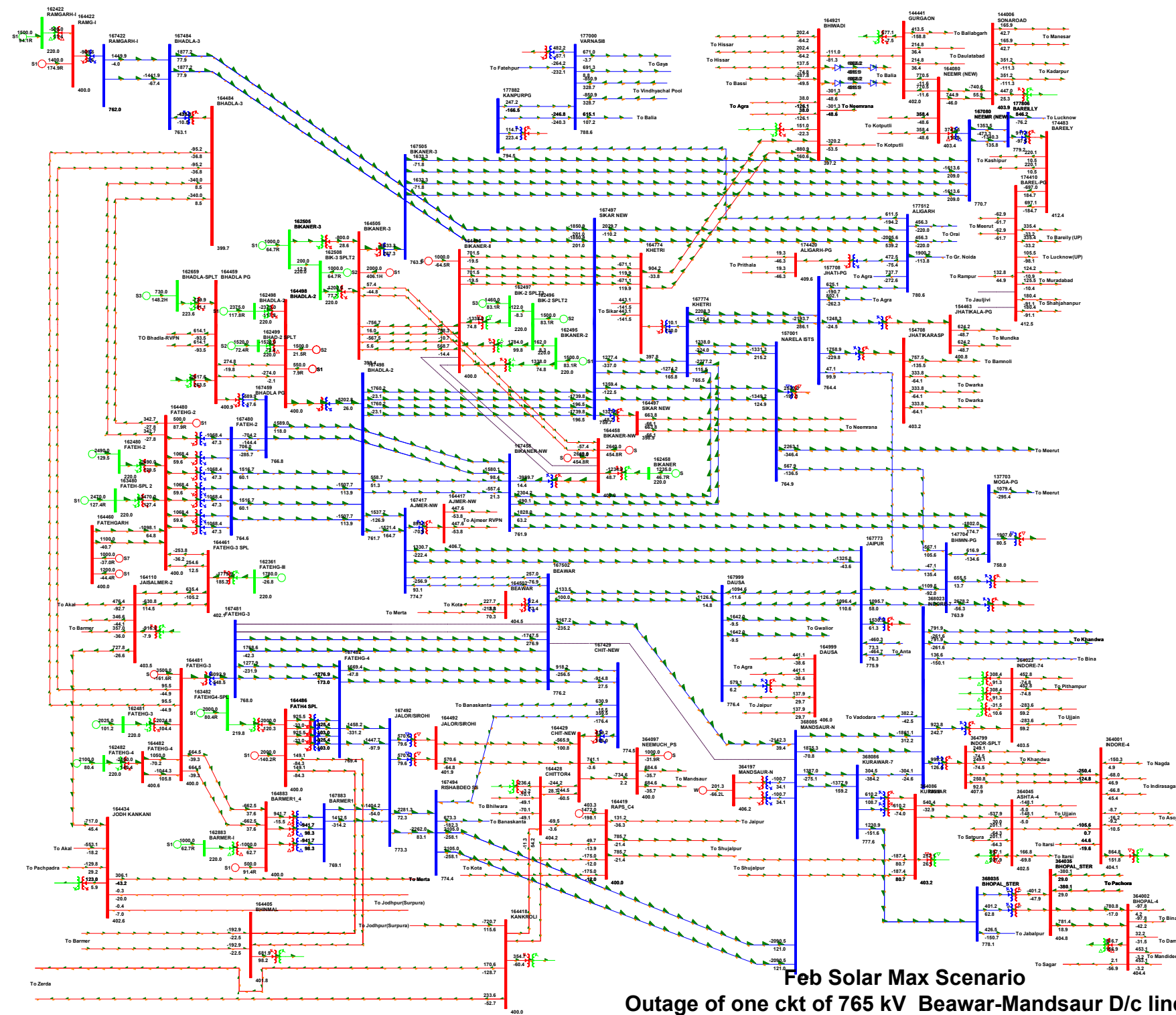
Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



Feb Solar Max Scenario

Outage of one ckt of 765 kV Rishabhdeo-Mandsaur D/c line

Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)



Transmission System for Rajasthan REZ Phase-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex)

